

THE (p, q) -ANALOGUES OF BI-STIRLING EULERIAN POLYNOMIALS VIA CONTINUED FRACTIONS

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ABSTRACT. We establish Jacobi-type continued fraction expansions for (p, q) -analogues of the bi-Stirling Eulerian polynomials, which refine classical Eulerian statistics on permutations. These polynomials admit equivalent definitions in terms of descents or excedances. The parameters p and q encode refined permutation statistics, including $(31-2, 2-13)$ and $(\text{cros}, \text{nest})$. Our approach is based on weighted Motzkin path models arising from variants of the bijections of Françon–Viennot and Foata–Zeilberger. As applications, we obtain new results on total positivity and γ -positivity for several families of enumerative polynomials. Our results unify and extend a number of recent works in the literature.

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1. INTRODUCTION

The Eulerian polynomials are fundamental objects in enumerative combinatorics, as they encode the distribution of ascents and descents over permutations. Over the years, numerous refinements and generalizations have been introduced, notably the bi-Stirling Eulerian polynomials, which incorporate record-type statistics, and the (p, q) -Eulerian polynomials, which refine Eulerian statistics via generalized permutation patterns. The aim of the present paper is to combine these two directions by introducing (p, q) -analogues of the bi-Stirling-Eulerian polynomials.

We now recall the necessary definitions and notation concerning permutation statistics and Eulerian-type polynomials. For a finite subset E of $\mathbb{N}$, let $\mathfrak{S}_E$ denote the set of all permutations of E , and let $\mathfrak{S}_n$ be the symmetric group on $[n] := \{1, 2, \dots, n\}$. For $\sigma \in \mathfrak{S}_n$, regarded as the word $\sigma = \sigma_1\sigma_2 \cdots \sigma_n$, where $\sigma_i = \sigma(i)$ for $i \in [n]$, an index $1 \leq i \leq n-1$ is called

- a *descent* (**des**) if $\sigma_i > \sigma_{i+1}$;
- an *ascent* (**asc**) if $\sigma_i < \sigma_{i+1}$.

Furthermore, a letter σ_i is said to be

- a *left-to-right maximum* (**lmax**) if $\sigma_i > \sigma_j$ for every $j < i$;
- a *right-to-left maximum* (**rmax**) if $\sigma_i > \sigma_j$ for every $j > i$.

Carlitz and Scoville [CS74] introduced the generalized Eulerian polynomials as the enumerative polynomials of the symmetric group $\mathfrak{S}_n$ with respect to ascent, descent, and record statistics, namely

$$A_n(x, y \mid \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (1.1)$$

Note that $\text{asc}(\sigma) + \text{des}(\sigma) = n$ for $\sigma \in \mathfrak{S}_{n+1}$. Motivated by their combinatorial interpretation [XZ26], we refer to these polynomials as the *bi-Stirling Eulerian polynomials*, although they are termed the (α, β) -Eulerian polynomials in [Ji25].

Recently, Ji [Ji25] introduced a refinement of $A_n(x, y \mid \alpha, \beta)$, defined by

$$A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (1.2)$$

In the refinement (1.2), the four parameters u_1, u_2, u_3, u_4 distinguish valleys, peaks, double ascents, and double descents, while the parameters α and β record left-to-right and right-to-left maxima. This refined description will later be incorporated into a single weight function $\mathbf{wt}(\sigma)$, together with additional statistics, including a D-statistic $D(\sigma)$ naturally arising from the Françon–Viennot bijection.

For $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, with the convention $\sigma_0 = \sigma_{n+1} = 0$, a letter σ_i with $i \in [n]$ is called a

- *valley* (**val**) if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$;
- *peak* (**pk**) if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- *double ascent* (**da**) if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;
- *double descent* (**dd**) if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$.

These statistics satisfy

$$\text{asc} = \text{val} + \text{da}, \quad \text{des} = \text{val} + \text{dd}, \quad \text{val} = \text{pk} - 1. \quad (1.3)$$

The exponential generating function (EGF) of $A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta)$ was obtained by Ji [Ji25, Theorem 1.4]. As observed in [XZ26], Ji's EGF formula can be written in the following form

$$\sum_{n \geq 0} A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta) \frac{z^n}{n!} = e^{(\alpha u_3 + \beta u_4)z} \left(\frac{x - y}{xe^{yz} - ye^{xz}} \right)^{\alpha + \beta}, \quad (1.4)$$

where the parameters x and y are related by $x + y = u_3 + u_4$ and $xy = u_1 u_2$. The sequel [XZ24] established a further generalization (1.4) and derived a J -fraction expansion for the ordinary generating function. In particular, building on earlier work of Zeng [Zen93], we obtained the continued fraction formula

$$\sum_{n=0}^{\infty} A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta) z^n = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{1 - b_2 z - \dots}}}, \quad (1.5a)$$

where $b_0 = \alpha u_3 + \beta u_4$,

$$b_k = k(u_3 + u_4) + (\alpha u_3 + \beta u_4), \quad \lambda_k = k(k - 1 + \alpha + \beta)u_1 u_2 \quad (k \geq 1). \quad (1.5b)$$

In this paper, we will generalize the above continued fraction formula by using the combinatorial theory of continued fractions developed by Flajolet [Fla80] and Viennot [Vie83], together with variants of the bijections of Françon–Viennot [FV79] and Foata–Zeilberger [FZ90] between permutations and weighted Motzkin paths. This approach allows us to derive explicit Jacobi-type continued fraction expansions and to obtain new consequences concerning positivity properties of the associated enumerative polynomials.

Our approach is based on continued fractions arising from weighted Motzkin path models. These models provide transparent combinatorial interpretations and lead naturally to explicit Jacobi-type continued fraction expansions for the ordinary generating functions under consideration.

More generally, our work fits naturally into the broad framework connecting permutation statistics, moment sequences, orthogonal polynomials, and continued fractions. In many situations, continued fractions arising from permutation statistics admit an interpretation as moment generating functions of suitable measures, thereby providing a bridge between enumerative combinatorics and the analytic theory of moments. This perspective has been developed, among others, by Blitvić and Steingrímsson [BS21] in their study of permutations, moments, and measures, and more recently by Sokal and Zeng [SZ22] through the introduction of multivariate master polynomials for permutations, set partitions, and perfect matchings together with their continued-fraction expansions. Building on the classical Flajolet–Viennot theory, Pétréolle, Sokal, and Zhu [PSZ23] extended Jacobi-type continued fractions to lattice paths and branched continued fractions, with applications to coefficient-wise Hankel-total positivity.

In this context, weighted Motzkin paths and Laguerre histories provide a combinatorial realization of moment sequences, where the step weights encode refined permutation statistics. The Foata–Zeilberger correspondence and its variants therefore furnish a natural mechanism for translating permutation statistics into continued-fraction expansions, and will play a central role in the bijections and J -fraction formulas developed in this paper. The present work offers further evidence for the relevance of this framework by introducing new

families of Eulerian-type polynomials whose generating functions admit explicit Jacobi-type continued-fraction expansions.

2. MAIN RESULTS

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is said to be a

- *left-to-right-maximum-peak* (**lmaxpk**) if σ_i is a left-to-right maximum and also a peak;
- *right-to-left-maximum-peak* (**rmaxpk**) if σ_i is a right-to-left maximum and also a peak;
- *left-to-right-maximum-double-ascent* (**lmaxda**) if σ_i is a left-to-right maximum and also a double ascent;
- *right-to-left-maximum-double-descent* (**rmaxdd**) if σ_i is a right-to-left maximum and also a double descent.

Example 1. Let $\sigma = 27159108436 \in \mathfrak{S}_{10}$, we have

$$\begin{aligned} \text{pk}(\sigma) &= |\{6, 7, 10\}| = 3, \quad \text{val}(\sigma) = |\{1, 3\}| = 2, \quad \text{da}(\sigma) = |\{2, 5, 9\}| = 3, \quad \text{dd}(\sigma) = |\{4, 8\}| = 2, \\ \text{lmax}(\sigma) &= |\{2, 7, 9, 10\}| = 4, \quad \text{rmax}(\sigma) = |\{6, 8, 10\}| = 3, \\ \text{lmaxda}(\sigma) &= |\{2, 9\}| = 2, \quad \text{rmaxdd}(\sigma) = |\{8\}| = 1, \\ \text{lmaxpk}(\sigma) &= |\{7, 10\}| = 2, \quad \text{and} \quad \text{rmaxpk}(\sigma) = |\{6, 10\}| = 2. \end{aligned}$$

As a follow-up to [Ji25] the authors [XZ24] generalized (1.2) with three additional variables as follows:

$$A_n(\mathbf{u}, f, g, t \mid \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma), \quad (2.1a)$$

where $\mathbf{u} = (u_1, u_2, u_3, u_4)$ and

$$\begin{aligned} \mathbf{wt}(\sigma) &:= (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} \\ &\quad \times f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)}. \end{aligned} \quad (2.1b)$$

Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = 0$. Denote by $\text{Val}(\sigma)$ and $\text{Pk}(\sigma)$ the sets of valleys and peaks of σ , respectively. We define the D statistic of σ by

$$D(\sigma) := \sum_{i \in \text{Pk}(\sigma) \setminus \{n\}} i - \sum_{j \in \text{Val}(\sigma)} j. \quad (2.2)$$

This statistic corresponds to the area under the associated Motzkin path via the Françon–Viennot bijection, see Lemma 3.4. Note that a shifted version of D statistic appeared in [Fra92].

Example 2. Let $\sigma = 42157368 \in \mathfrak{S}_8$. Then $\text{Val}(\sigma) = \{1, 3\}$, $\text{Pk}(\sigma) = \{4, 7, 8\}$, and hence $D(\sigma) = 4 + 7 - 1 - 3 = 7$.

The notion of generalized (dashed) permutation patterns was introduced by Babson and Steingrímsson [BS00]. Recall that the pattern $31-2$ (resp. $2-13$) occurs in a permutation σ if there exist indices $i < j$ such that the entries in positions i and $i+1$ (resp. j and $j+1$) are consecutive in σ , and

$$\sigma(i) > \sigma(j) > \sigma(i+1) \quad (\text{resp. } \sigma(j) < \sigma(i) < \sigma(j+1)).$$

We denote by $(31-2)(\sigma)$ (resp. $(2-13)(\sigma)$) the number of occurrences of the pattern $31-2$ (resp. $2-13$) in σ ; see [SZ12]. Accordingly, we set

$$31-2(\sigma) := (31-2)(\sigma) \quad \text{and} \quad 2-13(\sigma) := (2-13)(\sigma).$$

Similarly we define the pattern $2-31$ and the number $(2-31)(\sigma)$.

Example 3. Let $\sigma = 4736215 \in \mathfrak{S}_7$. Then

$$\begin{aligned} (31-2)(\sigma) &= |\{73-6, 73-5, 62-5\}| = 3, \\ (2-31)(\sigma) &= |\{4-73, 4-62, 3-62\}| = 3, \\ (2-13)(\sigma) &= |\{4-36, 4-15, 3-15, 2-15\}| = 4. \end{aligned}$$

Shin and one of us [SZ12] proved a continued fraction formula for the ordinary generating function of the following (p, q) -analogue of generalized Eulerian polynomials:

$$A_n(u_1, u_2, u_3, u_4; p, q) = \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} p^{2-13(\sigma)} q^{31-2(\sigma)} \quad (2.3)$$

and derived the γ -positivity formula using combinatorics of continued fractions. Combining (1.2) and (2.3), we define the (p, q) -analogue of the generalized Eulerian polynomials as follows:

$$A_n(\mathbf{u}, f, g, h, t \mid \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma) h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}. \quad (2.4)$$

We define the generalized (p, q) -analogue of a positive integer $k \in \mathbb{N}^*$, with auxiliary parameters α and β , as follows:

$$[k; \alpha, \beta]_{p,q} := (\alpha - 1)p^{k-1} + (\beta - 1)q^{k-1} + \frac{p^k - q^k}{p - q}. \quad (2.5a)$$

Note that

$$[1; \alpha, \beta]_{p,q} = \alpha + \beta - 1,$$

and

$$[k]_{p,q} := [k; 1, 1]_{p,q} = \frac{p^k - q^k}{p - q}. \quad (2.5b)$$

Let $\mathbf{b} := (b_k)_{k \geq 0}$ and $\boldsymbol{\lambda} := (\lambda_k)_{k \geq 1}$ be elements of a commutative ring. We define the Jacobi-type continued fraction $\mathcal{J}(t; \mathbf{b}, \boldsymbol{\lambda})$ as a formal power series by

$$\mathcal{J}(t; \mathbf{b}, \boldsymbol{\lambda}) := \frac{1}{1 - b_0 t - \frac{\lambda_1 t^2}{1 - b_1 t - \frac{\lambda_2 t^2}{1 - b_2 t - \ddots}}}. \quad (2.6)$$

Theorem 2.1. We have

$$\sum_{n=0}^{\infty} A_n(\mathbf{u}, f, g, h, t \mid \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.7a)$$

where

$$b_k = h^k \left(u_3 [k+1; 1, \alpha t]_{p,q} + u_4 [k+1; \beta t, 1]_{p,q} \right), \quad (2.7b)$$

$$\lambda_k = u_1 u_2 h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}. \quad (2.7c)$$

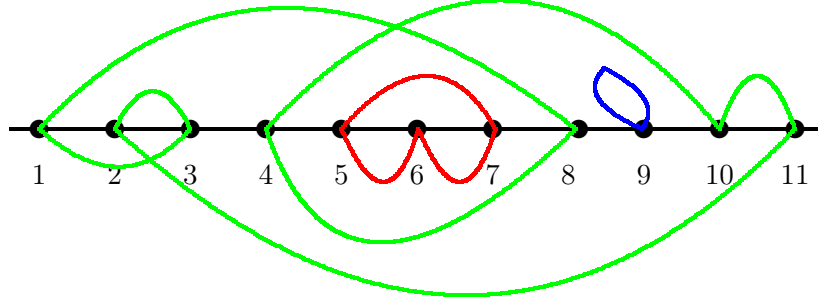


FIGURE 1. The arc-diagram of the permutation $\sigma = 8\,3\,1\,10\,7\,5\,6\,4\,9\,11\,2$, or as product of disjoint cycles $\sigma = (1, 8, 4, 10, 11, 2, 3) (5, 7, 6) (9)$.

Remark 2.1. The special case $f = g = h = t = p = q = 1$ of (2.7) is Eq. (1.5a). When $\alpha = \beta = h = 1$, other generalizations of the above continued fraction were considered in [SS96; BS21; SZ22].

Let $\sigma = \sigma_1\sigma_2\cdots\sigma_n \in \mathfrak{S}_n$. An *inversion* of σ is a pair of indices (i, j) such that $1 \leq i < j \leq n$ and $\sigma_i > \sigma_j$. The *inversion number* of σ , denoted by $\text{inv}(\sigma)$, is the total number of inversions in σ . Petersen and Tenner [PT15] introduced a statistic called *depth* for Coxeter groups, defined in terms of factorizations of group elements into reflections. In the case of the symmetric group $\mathfrak{S}_n$, the reflections are the transpositions $(i\,j)$ with $1 \leq i < j \leq n$. The *depth* of a permutation $\sigma \in \mathfrak{S}_n$, denoted by $\text{depth}(\sigma)$, is defined as

$$\text{depth}(\sigma) = \min \left\{ \sum_{r=1}^k (j_r - i_r) : \sigma = (i_1\,j_1)(i_2\,j_2)\cdots(i_k\,j_k) \right\}. \quad (2.8)$$

They obtained an alternative formula

$$\text{depth}(\sigma) = \sum_{\sigma(i) > i} (\sigma(i) - i). \quad (2.9)$$

Thus, this statistic coincides with the statistic *Edif* over permutations in [CSZ97] and corresponds to the area under the Motzkin path via the bijection of Foata-Zeilberger, see Lemma 3.6.

Corteel [Cor07] defined the *crossing* and *nesting* numbers of $\sigma \in \mathfrak{S}_n$ as follows:

$$\text{cros}(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j \leq \sigma_i < \sigma_j) \vee (i > j > \sigma_i > \sigma_j)\}; \quad (2.10a)$$

$$\text{nest}(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j \leq \sigma_j < \sigma_i) \vee (i > j > \sigma_j > \sigma_i)\}. \quad (2.10b)$$

These statistics are better illustrated by drawing an arc-diagram as follows: put the numbers from 1 to n on a line and draw an edge from i to σ_i above the line if i is a weak excedance, and below the line otherwise.

Example 4. Let $\sigma = 8\,3\,1\,10\,7\,5\,6\,4\,9\,11\,2 \in \mathfrak{S}_{11}$. Then we have $\text{nest}(\sigma) = 9$ and $\text{cros}(\sigma) = 3$; see Fig. 1.

For $\sigma = \sigma(1)\cdots\sigma(n) \in \mathfrak{S}_n$, we say that an index $i \in [n]$ is a

- *fixed point* (**fix**) of σ if $i = \sigma(i)$;
- *cycle peak* (**cpk**) of σ if $\sigma^{-1}(i) < i > \sigma(i)$;
- *cycle valley* (**cval**) of σ if $\sigma^{-1}(i) > i < \sigma(i)$;

- *cycle double ascent* (**cda**) of σ if $\sigma^{-1}(i) < i < \sigma(i)$;
- *cycle double descent* (**cdd**) of σ if $\sigma^{-1}(i) > i > \sigma(i)$.

Note that $\text{cpk}(\sigma) = \text{cval}(\sigma)$. A letter $\sigma_i := \sigma(i)$ is said to be a right-to-left minimum (**rmin**) of σ if $\sigma_j > \sigma_i$ and $n \geq j > i \geq 1$. Define the general cycle version of Eulerian polynomials

$$C_n(\mathbf{u}, t, s, h \mid \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} \\ \times h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)} \beta^{\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}. \quad (2.11a)$$

Theorem 2.2. *We have*

$$1 + \sum_{n \geq 1} C_n(\mathbf{u}, t, s, h \mid \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.11b)$$

where $b_0 = t\alpha\beta$, and

$$b_k = (sh)^k \left(tq^k s^k + u_4[k; \beta, 1]_{p,qs} + u_3 p[k; \alpha, 1]_{p,qs} \right), \quad (2.11c)$$

$$\lambda_k = u_1 u_2 (sh)^{2k-1} [k; \alpha, 1]_{p,qs} [k; \beta, 1]_{p,qs}. \quad (2.11d)$$

Remark 2.2. *Theorem 2.2 specializes to Theorem 6.5 of [MV94] when $\alpha = p = q = h = t = 1$, after a suitable change of variables. When $u_1 = u_4 = s = h = \alpha = \beta = 1$ and $u_2 = u_3 = t = x$, Theorem 2.2 reduces to Eq. (1) in [Cor07].*

Remark 2.3. *Petersen [Pet] conjectured the special case of the above J -fraction expansion corresponding to the pair of statistics $(\text{inv}, \text{depth})$; this conjecture was recently proved in [EFL24; SS24]. In fact, this J -fraction also follows from Theorem 10 or Corollary 11 of [CSZ97] after the successive substitutions $p \rightarrow q$ and then $q \rightarrow zq$.*

Remark 2.4. *In Theorem 2.2, applying the following substitutions, respectively,*

- $\mathbf{u} = (x, 1, x, 1)$ and $(\alpha, \beta, p, q) = (1, 1, 1, 1)$,
- $\mathbf{u} = (x, 1, x, 1)$ and $(t, p, q, \beta) = (tx, 1, 1, 1)$,
- $\mathbf{u} = (x, 1, x, 1)$ and $(t, p, q, \alpha) = (1, 1, 1, 1)$,

we recover Theorem 1.1, (i) and (ii) of Theorem 1.3 in [EFL26].

Definition 2.1. *Let $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. If $i \in [n-1]$ is a descent, then σ_i is called a descent-top, and σ_{i+1} is called a descent-bottom. For $i \in [n]$, the entry σ_i is called a nondescent or a nondescent-top (resp. nondescent-bottom) of σ , if it is not a descent-top (resp. descent-bottom) of σ . The number of nondescents of σ is $\text{ndes}(\sigma) := n - \text{des}(\sigma)$.*

Let $\sigma \in \mathfrak{S}_n$. We denote the sum of the nondescent-tops (resp. nondescent-bottoms) of σ by $\text{NDtop}(\sigma)$ (resp. $\text{NDbot}(\sigma)$) and define the *nondescent difference* of σ (see [SZ10]) by

$$\text{NDdif}(\sigma) := \text{NDbot}(\sigma) - \text{NDtop}(\sigma). \quad (2.12)$$

If we impose the boundary condition $\sigma_0 = 0$ and $\sigma_{n+1} = n+1$, then each nondescent-top (resp. nondescent-bottom) of $\sigma \in \mathfrak{S}_n$ is either a valley or a double ascent (resp. a peak or a double ascent), it follows that

$$\text{NDdif}(\sigma) = \sum_{i \in \text{Pk}'(\sigma)} i - \sum_{j \in \text{Val}'(\sigma)} j, \quad (2.13)$$

where $\text{Pk}'(\sigma)$ and $\text{Val}'(\sigma)$ denote the sets of peaks and valleys of σ , respectively, with boundary condition $\sigma_0 = 0$ and $\sigma_{n+1} = n+1$.

Example 5. Let $\sigma = 42157368 \in \mathfrak{S}_8$. We have

$$\text{NDtop}(\sigma) = 1 + 3 + 5 + 6 + 8 = 23,$$

$$\text{NDbot}(\sigma) = 4 + 5 + 6 + 7 + 8 = 30.$$

Then $\text{NDdif}(\sigma) = 7$. By (2.13), we have $\text{NDdif}(\sigma) = (4 + 7) - (1 + 3) = 7$.

Definition 2.2. Let $\sigma = \sigma_1\sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. A letter $\sigma_i \in [n]$ is called a

- *foremaximum* of σ if σ_i is a nondescent top and also a left-to-right maximum in σ ;
- *postminimum* of σ if σ_i is a nondescent bottom and also a right-to-left minimum in σ .

The number of foremaxima (resp. postminima) of σ is denoted by $\text{fmax}(\sigma)$ (resp. $\text{pmin}(\sigma)$).

Example 6. For $\sigma = 42157368 \in \mathfrak{S}_8$, we have $\text{fmax}(\sigma) = |\{5, 8\}| = 2$ and $\text{pmin}(\sigma) = |\{6, 8\}| = 2$.

Recall [SZ10] that for $\sigma \in \mathfrak{S}_n$ the statistics MAD and MADL are defined by

$$\text{MAD}(\sigma) := \text{des}(\sigma) + (31-2)(\sigma) + 2(2-31)(\sigma); \quad (2.14a)$$

$$\text{MADL}(\sigma) := \text{des}(\sigma) + (2-31)(\sigma) + 2(31-2)(\sigma). \quad (2.14b)$$

For $\sigma \in \mathfrak{S}_n$, an index $i \in [n]$ is called

- an *excedance* (**exc**) of σ if $i < \sigma(i)$,
- a *weak excedance* (**wex**) of σ if $i \leq \sigma(i)$,
- a *drop* (**drop**) if $\sigma(i) < i$.

Clarke, Steingrímsson and Zeng [CSZ97] constructed a bijection Φ_{CSZ} on $\mathfrak{S}_n$ to characterize the composition $\Phi_{FZ}^{-1} \circ \Phi_{FV}$. Later, Corteel [Cor07] and Shin–Zeng [SZ10] introduced two variants of the bijection Φ_{CSZ} on $\mathfrak{S}_n$, denoted by Φ_C and Φ_{SZ} , respectively. These bijections are known to send the generalized pattern statistic

$$(31-2, 2-31)$$

to the pairs (cros, nest) and (nest, cros) on $\mathfrak{S}_n$. Beyond this correspondence, they admit a natural interpretation in terms of the weighted path model underlying our continued-fraction expansions. In particular, the following result shows that Φ_C and Φ_{SZ} preserve additional structural features compatible with the Motzkin-path encoding and the associated Jacobi-type continued fractions. Related aspects of these constructions have recently been investigated in [HMZ20; CF23; CFZ25].

Theorem 2.3. The mappings Φ_C and Φ_{SZ} are bijections on $\mathfrak{S}_n$ such that for all $\sigma \in \mathfrak{S}_n$, we have

$$(\text{ndes}, \text{pmin}, 31-2, 2-31, \text{MADL}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{nest}, \text{cros}, \text{inv}, \text{depth}) \Phi_C(\sigma); \quad (2.15)$$

$$(\text{ndes}, \text{fmax}, 31-2, 2-31, \text{MAD}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{cros}, \text{nest}, \text{inv}, \text{depth}) \Phi_{SZ}(\sigma). \quad (2.16)$$

Moreover, if $\Phi^* := \Phi_{SZ} \circ \Phi_C^{-1}$, then

$$(\text{wex}, \text{nest}, \text{cros}, \text{depth}) \sigma = (\text{wex}, \text{cros}, \text{nest}, \text{depth}) \Phi^*(\sigma). \quad (2.17)$$

Any polynomial with real coefficients $h(x) = \sum_{i=0}^n h_i x^i$ satisfying the symmetry condition $h_i = h_{n-i}$ can be written uniquely in the form

$$h(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k x^k (1+x)^{n-2k}.$$

The coefficients γ_k are called the γ -coefficients of $h(x)$. If all γ_k are nonnegative, then $h(x)$ is said to be γ -positive. We refer the reader to the general overview of γ -positivity phenomena in [Ath18]. We now consider two specializations of the polynomials $A_n(\mathbf{u}, f, g, h, t \mid \alpha, \beta; p, q)$:

$$A_n(x, f, g, h \mid p, q) := A_n((1, x, 1, x), f, g, h, 1 \mid 1, 1; p, q); \quad (2.18a)$$

$$B_n(x, f, g, h \mid t, \alpha) := A_n((1, x, 1, x), f, g, h, t \mid \alpha, \alpha; 1, 1). \quad (2.18b)$$

By (2.4), we obtain

$$A_n(x, f, g, h \mid p, q) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{w}(\sigma) p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.18c)$$

$$B_n(x, f, g, h \mid t, \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{w}(\sigma) t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}, \quad (2.18d)$$

where the weight $\mathbf{w}(\sigma)$ is defined by

$$\mathbf{w}(\sigma) = x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma) - 1} g^{\text{rmaxpk}(\sigma) - 1} h^{\text{D}(\sigma)}. \quad (2.18e)$$

By the J -fraction expansion of Theorem 2.1, we show that the polynomials $A_n(x, f, g, h \mid p, q)$ and $B_n(x, f, g, h \mid t, \alpha)$ admit natural γ -positive decompositions. The next result provides an explicit combinatorial interpretation of the γ -coefficients in terms of permutations without double descents, refined by generalized pattern statistics.

Theorem 2.4. *For $n \geq 0$, the polynomial $A_n(x, f, g, h \mid p, q)$ is γ -positive and can be expanded as*

$$A_n(x, f, g, h \mid p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(f) x^k (1+x)^{n-2k}, \quad (2.19)$$

where

$$\gamma_{n,k}^a(f) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma) - 1} g^{\text{rmaxpk}(\sigma) - 1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.20)$$

and

$$\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}.$$

Remark 2.5. *When $f = g = h = 1$, this expansion reduces to a classical γ -positive decomposition of Eulerian-type polynomials. An equivalent formulation of (2.19) was obtained by Brändén [Brä08, Eq. (5.1)] via the modified Foata–Strehl action. The present approach derives the γ -coefficients directly from the continued-fraction structure.*

Theorem 2.5. *For $n \geq 0$, we have*

$$B_n(x, f, g, h \mid t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(f) x^k (1+x)^{n-2k}, \quad (2.21)$$

where

$$\gamma_{n,k}^b(f) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma) - 1} g^{\text{rmaxpk}(\sigma) - 1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}, \quad (2.22)$$

with $\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}$.

Remark 2.6. *When $f = g = h = 1$, we recover (3.28c) of Proposition 3.10 in [XZ24].*

Postnikov, Reiner, and Williams [PRW08] proved that the h -polynomials of dual stellohedra has the combinatorial interpretation

$$\tilde{A}_n(x) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)},$$

where PRW_n is the set of permutations in $\mathfrak{S}_n$ such that the first descent is at the letter n . For example,

$$\text{PRW}_2 = \{12, 21\} \quad \text{and} \quad \text{PRW}_3 = \{123, 132, 231, 312, 321\}.$$

Clearly, a permutation σ of $[n]$ is in PRW_n if and only if n is simultaneously the unique left-to-right maximum and a peak of σ , namely, $\text{lmaxpk}(\sigma) = 1$. By (2.18), we have

$$A_n(x, 0, g, h \mid p, q) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.23a)$$

$$B_n(x, 0, g, h \mid t, \alpha) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2}. \quad (2.23b)$$

The following result is an immediate consequence of Theorems 2.4 and 2.5.

Theorem 2.6. *We have*

$$A_n(x, 0, g, h \mid p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(0) x^k (1+x)^{n-2k}, \quad (2.23c)$$

$$B_n(x, 0, g, h \mid t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(0) x^k (1+x)^{n-2k}, \quad (2.23d)$$

where

$$\gamma_{n,k}^a(0) = \sum_{\sigma \in \text{PRW}_{n+1,k}^{\text{dd}=0}} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.23e)$$

$$\gamma_{n,k}^b(0) = \sum_{\sigma \in \text{PRW}_{n+1,k}^{\text{dd}=0}} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (2.23f)$$

with

$$\text{PRW}_{n,k}^{\text{dd}=0} := \{\sigma \in \text{PRW}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}. \quad (2.23g)$$

Remark 2.7. *Special cases of (2.23d) were obtained in [XZ24, Proposition 3.8] and [JL24, Theorem 2.1].*

Consider the two special cases of the polynomials $C_n(\mathbf{u}, t, s, h \mid \alpha, \beta; p, q)$, see (2.11a),

$$\begin{aligned} C_n^1(x, t, s, h \mid \alpha, p, q) &:= C_n((x, 1, x/p, 1), t(1+x), s, h \mid \alpha, \alpha; p, q) \\ &= \sum_{\sigma \in \mathfrak{S}_n} \mathbf{w}^c(\sigma) p^{\text{cros}(\sigma)-\text{cda}(\sigma)} q^{\text{nest}(\sigma)}, \end{aligned} \quad (2.24a)$$

and

$$\begin{aligned} C_n^2(x, t, s, h \mid \alpha, p, q) &:= C_n((x, p, x, p), t(1+x), s, h \mid \alpha, \alpha; p, q) \\ &= \sum_{\sigma \in \mathfrak{S}_n} \mathbf{w}^c(\sigma) p^{\text{cros}(\sigma)+\text{drop}(\sigma)} q^{\text{nest}(\sigma)}, \end{aligned} \quad (2.24b)$$

where

$$\mathbf{w}^c(\sigma) = x^{\text{exc}(\sigma)} (t(1+x))^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma)+\text{lmax}(\sigma)}. \quad (2.24c)$$

By the J -fraction of Theorem 2.2, we show that the polynomials $C_n^1(x, t, s, h \mid \alpha, p, q)$ and $C_n^2(x, t, s, h \mid \alpha, p, q)$ admit the following γ -expansions.

Theorem 2.7. *For $n \geq 0$, we have*

$$C_n^i(x, t, s, h \mid \alpha, p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^i(t) x^k (1+x)^{n-2k}, \quad (i = 1, 2) \quad (2.25)$$

where

$$\gamma_{n,k}^1(t) = \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0}} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)} \quad (2.26)$$

$$\gamma_{n,k}^2(t) = \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0}} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmin}(\sigma)} p^{\text{cros}(\sigma)+\text{drop}(\sigma)} q^{\text{nest}(\sigma)} \quad (2.27)$$

with

$$\mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0} := \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0 \text{ and } \text{exc}(\sigma) = k\}. \quad (2.28)$$

Remark 2.8. *The special case $\alpha = s = h = 1$ of Eq. (2.25) with $i = 2$ is equivalent to Eq. (4.14) of Theorem 4.8 in [LWZ19].*

The next corollary corresponds to the derangement specialization of the γ -positive decomposition arising from the J -fraction in Theorem 2.1. By setting the fixed-point parameter to zero, the continued-fraction structure naturally restricts to weighted derangements, yielding an explicit combinatorial interpretation of the γ -coefficients. Define

$$C_n^1(x, 0, s, h \mid \alpha, p, q) = \sum_{\sigma \in \mathcal{D}_n} x^{\text{exc}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmin}(\sigma)} p^{\text{cros}(\sigma)-\text{cda}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.29)$$

$$C_n^2(x, 0, s, h \mid \alpha, p, q) = \sum_{\sigma \in \mathcal{D}_n} x^{\text{exc}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmin}(\sigma)} p^{\text{cros}(\sigma)+\text{drop}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.30)$$

where $\mathcal{D}_n := \{\sigma \in \mathfrak{S}_n : \text{fix}(\sigma) = 0\}$ is the set of derangements of $[n]$.

Corollary 2.8. *For $n \geq 0$, we have*

$$C_n^i(x, 0, s, h \mid \alpha, p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^i(0) x^k (1+x)^{n-2k}, \quad (i = 1, 2) \quad (2.31)$$

where

$$\gamma_{n,k}^1(0) = \sum_{\sigma \in \mathcal{D}_{n,k}^0} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma)+\text{lmax}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.32)$$

$$\gamma_{n,k}^2(0) = \sum_{\sigma \in \mathcal{D}_{n,k}^0} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma)+\text{lmax}(\sigma)} p^{\text{cros}(\sigma)+\text{drop}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.33)$$

with $\mathcal{D}_{n,k}^0 := \{\sigma \in \mathcal{D}_n : \text{cda}(\sigma) = 0 \text{ and } \text{exc}(\sigma) = k\}$.

Remark 2.9. *When $h = \alpha = q = 1$, Eq. (2.31) reduces to Theorem 2 of [SZ16].*

The remainder of this paper is organized as follows. In Section 3, we begin with a detailed review of the Françon–Viennot bijection and the Foata–Zeilberger bijection between permutations and Laguerre histories, a class of weighted Motzkin paths. We then employ these bijections to prove Theorems 2.1 and 2.2 using Flajolet’s fundamental theorem on J -continued fractions.

In Section 4, we recall the bijections of Shin–Zeng and Corteel on permutations and use them to establish Theorem 2.3. Section 5 is devoted to the γ -positive expansions derived from the continued fractions of Theorems 2.1 and 2.2. In particular, we obtain γ -expansions for the (p, q) -analogues of the generalized Eulerian polynomials (Theorems 2.4, 2.5, and 2.7). We also present two distinct factorizations of the corresponding γ -coefficients together with combinatorial interpretations in terms of André permutations of types 1 and 2; see Theorems 5.5, 5.8, and 5.9.

In Section 6, we apply the Pan–Zeng group action to prove Theorem 5.5 and the Dong–Lin–Pan group action to prove Theorem 5.8. Section 7 investigates the total positivity of the (p, q) -analogues of generalized Eulerian polynomials via continued fractions. Finally, in Section 8, we answer a question of Eu et al. by constructing a bijection that makes explicit the symmetry between two variants of the nest and cross statistics. We also derive several compact closed-form expressions obtained by specializing one of the parameters in the associated continued fractions to -1 .

3. ENCODING PERMUTATIONS BY LAGUERRE HISTORIES

3.1. Preliminaries. A *Motzkin path* of length n is a lattice path $\{(x_i, y_i)\}_{0 \leq i \leq n}$ in the plane that

- starts at $(x_0, y_0) = (0, 0)$ and ends at $(x_n, y_n) = (n, 0)$;
- uses only the three types of steps

$$N = (1, 1) \text{ (North-East)}, \quad S = (1, -1) \text{ (South-East)}, \quad E = (1, 0) \text{ (East)};$$

- never goes below the x -axis, that is, the height $h_i := y_{i-1} \geq 0$ for all $i \in [n]$.

Let $\mathcal{M}_n$ denote the set of Motzkin paths of length n .

Following the classical Flajolet–Viennot approach, we consider a *weighted Motzkin-path model*. Each East step at height i is assigned a weight a_i , while each North-East (resp. South-East) step at height i receives a weight b_i (resp. c_i). The weight of a path $P \in \mathcal{M}_n$ is the product of the weights of its steps and is denoted by $\omega(P)$. The generating function of weighted Motzkin paths admits a Jacobi-type continued-fraction expansion, given by Flajolet’s fundamental lemma on continued fractions [Fla80].

Lemma 3.1. *We have*

$$\sum_{n \geq 0} \sum_{P \in \mathcal{M}_n} \omega(P) z^n = \frac{1}{1 - b_0 z - \frac{a_0 c_1 z^2}{1 - b_1 z - \frac{a_1 c_2 z^2}{\ddots}}}. \quad (3.1)$$

For a permutation $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$ and $k = \sigma_j$ with $j \in [n]$, the coordinate generalized pattern statistics are defined by

$$(31-2)_k(\sigma) = \#\{i : i+1 < j \text{ and } \sigma_{i+1} < k < \sigma_i\};$$

$$\begin{aligned} (2-13)_k(\sigma) &= \#\{i : j < i-1 \text{ and } \sigma_{i-1} < k < \sigma_i\}; \\ (2-31)_k(\sigma) &= \#\{i : j < i-1 \text{ and } \sigma_i < k < \sigma_{i-1}\}. \end{aligned} \quad (3.2)$$

Clearly, for $\sigma \in \mathfrak{S}_n$, the statistic associated with the generalized pattern 31-2 decomposes as

$$(31-2)(\sigma) = \sum_{k=1}^n (31-2)_k(\sigma).$$

Similarly,

$$(2-13)(\sigma) = \sum_{k=1}^n (2-13)_k(\sigma), \text{ and } (2-31)(\sigma) = \sum_{k=1}^n (2-31)_k(\sigma).$$

Example 7. Let $\sigma = 4723516 \in \mathfrak{S}_7$. We have

$$(31-2)_3(\sigma) = 1, (2-13)_4(\sigma) = 2 \text{ and } (2-31)_4(\sigma) = 2.$$

We can identify a Motzkin path of length n by a Motzkin word $w = w_1 \dots w_n$ such that $w_i = \text{N}$ (resp. S, E) if and only if the i th step is N (resp. S, E). Given a Motzkin path $w = w_1 w_2 \dots w_n \in \mathcal{M}_n$, the height h_i of the i th step is given by

$$h_i := h_i(w) = \#\{j < i : w_j = \text{N}\} - \#\{j < i : w_j = \text{S}\}. \quad (3.3)$$

3.1.1. Françon-Viennot bijection. A 2-Motzkin path is a Motzkin path whose east steps could come in two colors: red (E_1) or blue (E_2).

Definition 3.1 (Laguerre history). A Laguerre history of length n is a couple (w, p) , where $w = w_1 \dots w_n$ is a 2-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is a label sequence such that $0 \leq p_i \leq h_i$ for $1 \leq i \leq n$. Denote by $\mathcal{L}_n$ the set of Laguerre histories of length n .

We describe a variant of Françon-Viennot's bijection $\Psi_{FV} : \mathfrak{S}_{n+1} \rightarrow \mathcal{L}_n$ and its inverse, see [SZ12; Vie83]. For a permutation $\sigma \in \mathfrak{S}_{n+1}$, the Laguerre history $\Psi_{FV}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n$ is defined by

$$w_i = \begin{cases} \text{N (resp. S)} & \text{if } i \text{ is a valley (resp. peak) of } \sigma; \\ \text{E}_1 \text{ (resp. E}_2\text{)} & \text{if } i \text{ is a double descent (resp. double ascent) of } \sigma, \end{cases} \quad (3.4)$$

and $p_i = (31-2)_i(\sigma)$ for each $i \in [n]$.

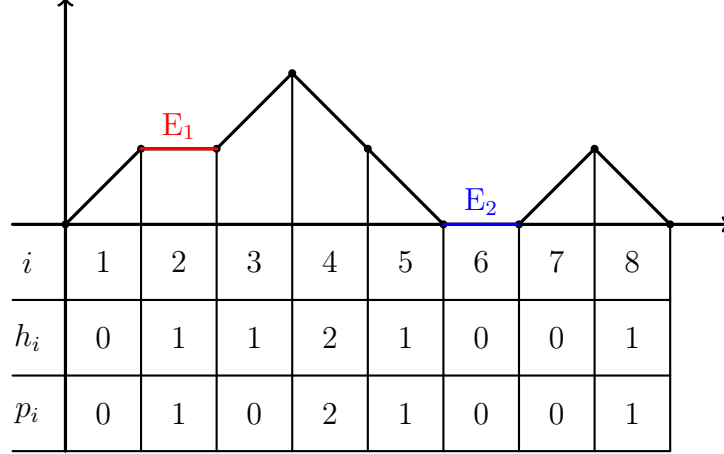
Inversely, starting from a Laguerre history $(w, p) \in \mathcal{L}_n$ one can recover the corresponding permutation σ by the following algorithm:

- Initialization: $\sigma = \diamond$;
- At the i th ($1 \leq i \leq n$) step of the algorithm, replace the $(p_i + 1)$ th $\diamond$ (from left to right) of σ by

$$\begin{cases} \diamond i \diamond & \text{if } w_i = \text{N}; \\ i \diamond & \text{if } w_i = \text{E}_2; \\ i & \text{if } w_i = \text{S}; \\ \diamond i & \text{if } w_i = \text{E}_1. \end{cases}$$

- The final permutation is obtained by putting $n + 1$ at the last remaining $\diamond$.

As a consequence, the cardinality of $\mathcal{L}_n$ is $(n + 1)!$.

FIGURE 2. The Laguerre history $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.

Example 8. Let $\sigma = 697835142 \in \mathfrak{S}_9$. We have

$$\Psi_{FV}(\sigma) = (\text{N}\text{E}_1\text{NSSE}_2\text{NS}, (0, 1, 0, 2, 1, 0, 0, 1)) \in \mathcal{L}_8,$$

see Fig. 2.

Inversely, we have

$$\begin{aligned} \sigma &= \diamond \rightarrow \diamond 1 \diamond \rightarrow \diamond 1 \diamond 2 \rightarrow \diamond 3 \diamond 1 \diamond 2 \rightarrow \diamond 3 \diamond 142 \rightarrow \diamond 35142 \rightarrow 6 \diamond 35142 \\ &\rightarrow 6 \diamond 7 \diamond 35142 \rightarrow 6 \diamond 7835142 \rightarrow 697835142. \end{aligned}$$

There is another more visual description of patterns $(31-2)_x$ and $(2-13)_x$ using the so-called x -decomposition; see [FV79]. For letter $x \in [n]$, the x -decomposition of σ is the unique factorization

$$\sigma = u_1 v_1 \cdots u_k v_k u_{k+1}, \quad k \geq 1, \quad (3.5)$$

in which the words u_i and v_j , except possibly u_1 and u_{k+1} , are nonempty. Moreover, each u_i ($1 \leq i \leq k+1$) consists solely of letters $< x$, while each v_j ($1 \leq j \leq k$) consists solely of letters $\geq x$. Assume that x lies inside v_j . Then every adjacent pair (v_i, u_{i+1}) with $i \leq j-1$ contributes to a $(31-2)_x(\sigma)$, while every adjacent pair (u_i, v_i) with $i \geq j$ contributes to a $(2-13)_x(\sigma)$. In Fig 3, we see that

$$(31-2)_x(\sigma) = j-1 \quad \text{and} \quad (2-13)_x(\sigma) = k-j.$$

Example 9. If $\sigma = 64195381027 \in \mathfrak{S}_{10}$ and $x = 5$. Then the 5-decomposition is

$$\sigma = u_1 v_1 u_2 v_2 u_3 v_3 u_4 v_4,$$

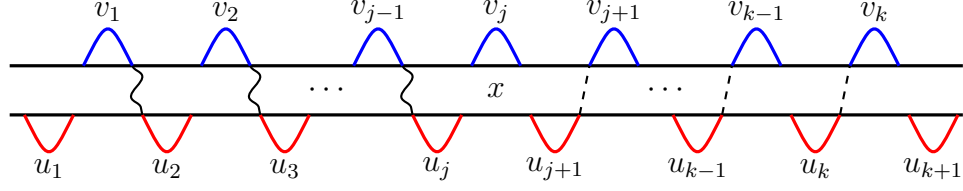
with

$$(u_1, u_2, u_3, u_4) = (\varepsilon, 41, 3, 2) \quad \text{and} \quad (v_1, v_2, v_3, v_4) = (6, 95, 810, 7).$$

Thus, we have $(31-2)_5(\sigma) = 1$ and $(2-13)_5(\sigma) = 2$.

Lemma 3.2. Given a permutation $\sigma \in \mathfrak{S}_{n+1}$ and the corresponding Laguerre history $\Psi_{FV}(\sigma) = (w, p) \in \mathcal{L}_n$. For any $x \in [n]$, if h_x is the height of the x th step of w , then

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = h_x. \quad (3.6)$$

FIGURE 3. The x -decomposition of σ .

Proof. Suppose that the x -decomposition of σ is

$$\sigma = u_1 v_1 \cdots u_k v_k u_{k+1}, \quad k \geq 1, \quad (3.7)$$

where x lies inside v_j . According to Definition 3.3 and the construction of Ψ_{FV} in (3.4), it follows that the height of the x th step of w is

$$h_x = \sum_{j=1}^{k+1} (\text{val}(u_j) - \text{pk}(u_j)),$$

where $\text{val}(u_j)$ (resp. $\text{pk}(u_j)$) denotes the number of valleys (resp. peaks) in u_j . It is straight-forward to verify that

- $\text{val}(u_j) = \text{pk}(u_j)$ for $j = 1, k+1$;
- $\text{val}(u_j) = \text{pk}(u_j) + 1$ for $2 \leq j \leq k$.

Hence we obtain $h_x(\sigma) = k - 1$. Every adjacent pair (v_i, u_{i+1}) with $i \leq j - 1$ contributes to a $(31-2)_x(\sigma)$, while every adjacent pair (u_i, v_i) with $i \geq j$ contributes to a $(2-13)_x(\sigma)$. Thus, h_x also can be calculated by using the patterns $(31-2)_x(\sigma)$ and $(2-13)_x(\sigma)$, that is (3.6).

Now we prove Eq. (3.6) by induction for all $x \in [n]$. For $x = 1$, it is obvious that $(31-2)_1(\sigma) + (2-13)_1(\sigma) = 0 = h_1$. If $x > 1$, it suffices to show that

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (31-2)_{x-1}(\sigma) + (2-13)_{x-1}(\sigma) + \nu, \quad (3.8)$$

where $\nu \in \{0, 1, -1\}$. Assume that the $(x-1)$ -decomposition of σ is

$$\sigma = u_1 v_1 \cdots u_k v_k u_{k+1}, \quad k \geq 1,$$

and $(x-1)$ is in v_j . In view of this $(x-1)$ -decomposition, we see that $(31-2)_{x-1}(\sigma) = j - 1$, $(2-13)_{x-1}(\sigma) = k - j$ and $(31-2)_{x-1}(\sigma) + (2-13)_{x-1}(\sigma) = k - 1$.

- For case $\nu = 1$. If $(x-1)$ is a valley of σ , and x is also a valley,
 - then x is in v_l where $1 \leq l < j$ (resp. $j < l \leq k$). In this case, $(x-1)$ together with its immediate right (resp. left) neighbor contributes 1 to $(2-13)_x(\sigma)$ (resp. $(31-2)_x(\sigma)$). Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (l - 1) + (k - l) + 1 = k.$$

- Or x is in v_j . Suppose that $v_j = v_j^l(x-1)v_j^r$ where v_j^l and v_j^r are (contiguous) words consisting of entries strictly larger than $x-1$. Both v_j^l and v_j^r are nonempty. If $x \in v_j^l$, then $(x-1)$ together with its immediate right neighbor contributes 1 to $(2-13)_x(\sigma)$; if $x \in v_j^r$, then $x-1$ together with its immediate left neighbor contributes 1 to $(31-2)_x(\sigma)$. Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (j - 1) + (k - j) + 1 = k.$$

Analogous arguments hold in the cases when x is a peak, a double ascent, or a double descent of σ .

- For case $\nu = -1$. If $(x-1)$ is a peak of σ , then from the $(x-1)$ -decomposition we see that v_j contains only $(x-1)$. Assume that x is a valley of σ , and that x is in v_l where $1 \leq l < j$ (resp. $j < l \leq k$). In the x -decomposition of σ , components u_j , $v_j = x-1$ and u_{j+1} will form a new u -word. Therefore, compared with the $(x-1)$ -decomposition, the x -decomposition has one fewer u -word. Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (l-1) + (k-l) - 1 = k-2.$$

Analogous arguments hold in the cases when x is a peak, a double ascent, or a double descent of σ .

- For case $\nu = 0$. If $(x-1)$ is a double descent (resp. double ascent) of σ , then $(x-1)$ is the last (resp. first) element of v_j . Assume that x is a valley of σ and that x is in v_l where $1 \leq l \leq k$. In these cases, it is easy to check that there is no change in the total number of v_i in both $(x-1)$ -decomposition and x -decomposition of σ . Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (l-1) + (k-l) = k-1.$$

Analogous arguments hold in the cases when x is a peak, a double ascent, or a double descent of σ . Combining these cases, we obtain (3.8). $\square$

For a Motzkin path w , we define $\text{area}(w)$ to be the geometric area of the region between the x -axis and the Motzkin path w , denoted by $\text{area}(w)$. We have the following characterization for area .

Lemma 3.3. *Let a Motzkin path $w \in \mathcal{M}_n$. The area of w is given by*

$$\text{area}(w) = \sum_{i=1}^n h_i, \quad (3.9)$$

where h_i is the height of the i th step of w defined in (3.3).

Proof. We compute $\text{area}(w)$ by the step heights of the Motzkin path $w \in \mathcal{M}_n$. The contribution of the i th step equals the area of the trapezoid of width 1 whose vertical sides have lengths h_i and $h_i + \Delta_i$, where $\Delta_i \in \{1, 0, -1\}$ is the vertical increment of that step. It is clear that the i th step contributes $h_i + \frac{\Delta_i}{2}$ to $\text{area}(w)$, therefore

$$\text{area}(w) = \sum_{i=1}^n \left(h_i + \frac{\Delta_i}{2} \right) = \sum_{i=1}^n h_i + \frac{1}{2} \sum_{i=1}^n \Delta_i.$$

Since w starts and ends on the x -axis, its total vertical increment is $\sum_{i=1}^n \Delta_i = 0$, and the above equation yields (3.9). $\square$

Lemma 3.4. *For $\sigma \in \mathfrak{S}_{n+1}$, let $\Psi_{FV}(\sigma) := (w, p) \in \mathcal{L}_n$ be the corresponding Laguerre history. Then*

$$D(\sigma) = \text{area}(w). \quad (3.10)$$

Proof. Let $w = w_1 \cdots w_n \in \mathcal{M}_n$. According to the construction of Ψ_{FV} in (3.4), we see that

$$D(\sigma) = \sum_{w_i=S} i - \sum_{w_j=N} j. \quad (3.11)$$

For each South-East step $w_i = S$, match it with the unique North-East step $w_j = N$ with $j < i$ such that the path between steps w_j and w_i stays strictly above height h_j , and $h_i = h_j + 1$. The matched pair of steps $w_i = S$ and $w_j = N$ determines a horizontal strip of area $i - j$ lying within the Motzkin path. Summing over all matched pairs gives

$$\text{area}(w) = \sum_{w_i=S} i - \sum_{w_j=N} j. \quad (3.12)$$

Combining this with Eq. (3.11) yields (3.10). $\square$

Example 10. Let $\sigma = 697835142 \in \mathfrak{S}_9$. The corresponding Motzkin path is $w = N\mathbf{E}_1\text{NSSSE}_2\text{NS}$; see Fig. 2. We have

$$\begin{aligned} D(\sigma) &= (8 + 5 + 4) - (7 + 3 + 1) = 6, \\ \text{area}(w) &= \sum_{i=1}^8 h_i = 4 \cdot 1 + 2 = 6. \end{aligned}$$

3.1.2. Foata-Zeilberger bijection. A *3-Motzkin path* is a Motzkin path whose east steps could come in three colors: red ($\mathbf{E}_1$), blue ($\mathbf{E}_2$) and green ($\mathbf{E}_3$).

Definition 3.2 (3-restricted Laguerre history). A *3-restricted Laguerre history* of length n is a couple (w, p) , where $w = w_1 \cdots w_n$ is a 3-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is a label sequence satisfying $1 \leq i \leq n$,

$$0 \leq p_i \leq \begin{cases} h_i, & \text{if } w_i \in \{N, \mathbf{E}_2, \mathbf{E}_3\}; \\ h_i - 1, & \text{if } w_i \in \{S, \mathbf{E}_1\}, \end{cases}$$

where h_i is the height of the i th step of w . Denote by $\mathcal{L}_n^*$ the set of 3-restricted Laguerre histories of length n .

For a permutation $\sigma \in \mathfrak{S}_n$, we describe Foata-Zeilberger's bijection $\Psi_{FZ}(\sigma) := (w, p) \in \mathcal{L}_n^*$ and its inverse. The associated 3-restricted Laguerre history

$$\Psi_{FZ}(\sigma) = ((w_1, \dots, w_n), (p_1, \dots, p_n)) \in \mathcal{L}_n^*$$

is defined as follows. The step w_i is given by

$$w_i = \begin{cases} N \text{ (resp. } S) & \text{if } i \text{ is a cycle valley (resp. cycle peak) of } \sigma; \\ \mathbf{E}_1 \text{ (resp. } \mathbf{E}_2) & \text{if } i \text{ is a cycle double descent (resp. cycle double ascent) of } \sigma; \\ \mathbf{E}_3 & \text{if } i \text{ is a fixed point of } \sigma, \end{cases} \quad (3.13)$$

and the integer p_i is defined by

$$p_i := \begin{cases} \#\{j < i : \sigma(j) > \sigma(i)\}, & \text{if } \sigma(i) \geq i; \\ \#\{j > i : \sigma(j) < \sigma(i)\}, & \text{if } \sigma(i) < i. \end{cases} \quad (3.14)$$

For a permutation $\sigma = \sigma(1) \cdots \sigma(n) \in \mathfrak{S}_n$, we say that i (resp. σ_i) is the *excedance bottom* (resp. *top*) of σ if $i < \sigma(i)$, and that i (resp. σ_i) is the *nonexcedance top* (resp. *bottom*) of σ if $i \geq \sigma(i)$. Let $\text{Etop}(\sigma)$, $\text{Ebot}(\sigma)$, $\text{NEtop}(\sigma)$ and $\text{NEbot}(\sigma)$ denote the sets of the excedance tops, excedance bottoms, nonexcedance tops and nonexcedance bottoms of σ , respectively. Let $\text{Cval}(\sigma)$, $\text{Cpk}(\sigma)$, $\text{Cda}(\sigma)$, $\text{Cdd}(\sigma)$ and $\text{Fix}(\sigma)$ be the sets of cycle valleys, cycle peaks, cycle double ascents, cycle double descents and fixed points of σ , respectively.

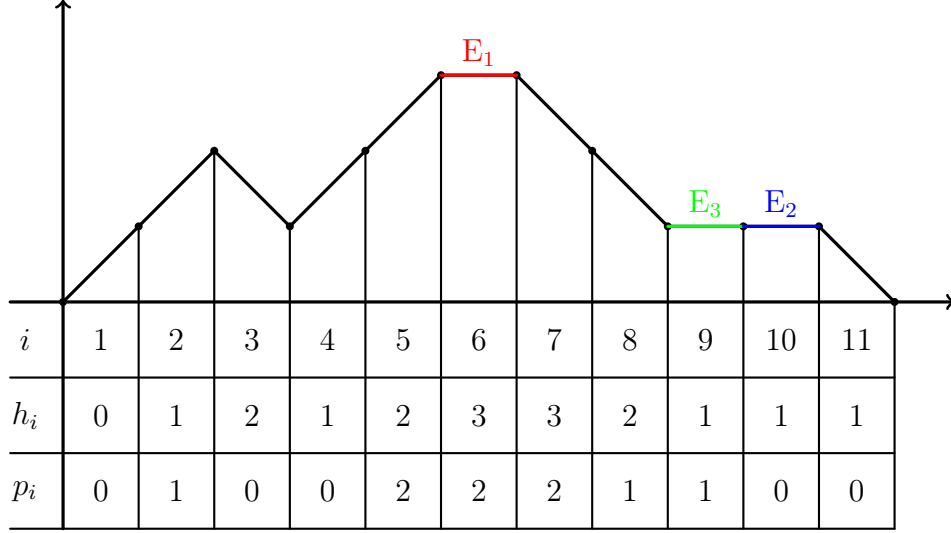


FIGURE 4. The 3-restricted Laguerre history $\Psi_{FZ}(\sigma)$ with $\sigma = 8311075649112 = (1, 8, 4, 10, 11, 2, 3)(5, 7, 6)(9)$.

It is easy to check that the following identities hold:

$$\text{Cpk}(\sigma) \cup \text{Cda}(\sigma) = \text{Etop}(\sigma), \quad \text{Cval}(\sigma) \cup \text{Cda}(\sigma) = \text{Ebot}(\sigma), \quad (3.15a)$$

$$\text{Cpk}(\sigma) \cup \text{Cdd}(\sigma) \cup \text{Fix}(\sigma) = \text{NEtop}(\sigma), \quad (3.15b)$$

$$\text{Cval}(\sigma) \cup \text{Cdd}(\sigma) \cup \text{Fix}(\sigma) = \text{NEbot}(\sigma). \quad (3.15c)$$

Definition 3.3. The left-to-right inversion table (or Lehmer code) of a permutation σ is the sequence $(e_1, e_2, \dots, e_n)$, where e_i denotes the number of entries to the left of position i that are bigger than σ_i , i.e.,

$$e_i = \#\{j < i : \sigma_j > \sigma_i\}. \quad (3.16)$$

Dually, the right-to-left inversion table of σ is the sequence $(f_1, \dots, f_n)$, where f_i denotes the number of entries to the right of position i that are smaller than σ_i , i.e.,

$$f_i = \#\{j > i : \sigma_j < \sigma_i\}. \quad (3.17)$$

The inverse algorithm Ψ_{FZ}^{-1} building a permutation σ from a 3-restricted Laguerre history $(w, p) \in \mathcal{L}_n^*$ can be described as follows:

According to the classification of steps in 3-restricted Laguerre history and Eq. (3.15), we first obtain the four sets $\text{Etop}(\sigma)$, $\text{Ebot}(\sigma)$, $\text{NEtop}(\sigma)$, and $\text{NEbot}(\sigma)$, where σ is the resulting permutation obtained at the end of the inverse bijection. We then construct two biwords,

$$\sigma_{\text{exc}} := \begin{pmatrix} \eta \\ \eta' \end{pmatrix} \text{ and } \sigma_{\text{nexc}} := \begin{pmatrix} \zeta \\ \zeta' \end{pmatrix},$$

where η and ζ are the subwords, written in increasing order, of the sets $\text{Ebot}(\sigma)$ and $\text{NEtop}(\sigma)$, respectively, and η' and ζ' are some subwords of $\text{Etop}(\sigma)$ and $\text{NEtop}(\sigma)$ obtained from the left-to-right and right-to-left tables with entries p_i , respectively. Then we recover the corresponding permutation $\Psi_{FZ}^{-1}((w, p))$.

As a consequence, the cardinality of $\mathcal{L}_n^*$ is $n!$.

Example 11. If $\sigma = 8311075649112 = (1, 8, 4, 10, 11, 2, 3)(5, 7, 6)(9) \in \mathfrak{S}_{11}$, then we have

$$\Psi_{FZ}(\sigma) = (\text{NNSNN}\mathbf{E}_1\text{SSE}_3\mathbf{E}_2\text{S}, (0, 1, 0, 0, 2, 2, 2, 1, 1, 0, 0)) \in \mathcal{L}_{11}^*.$$

Inversely, from the 3-restricted Laguerre history, we have

$$\begin{aligned} \text{Cval}(\sigma) &= \{1, 2, 4, 5\}, \text{Cpk}(\sigma) = \{3, 7, 8, 11\}, \\ \text{Cdd}(\sigma) &= \{6\}, \text{Cda}(\sigma) = \{10\} \text{ and } \text{Fix}(\sigma) = \{9\}. \end{aligned}$$

According to Eq. (3.15), we obtain

$$\begin{aligned} \text{Ebot}(\sigma) &= \{1, 2, 4, 5, 10\}, \text{Etop}(\sigma) = \{3, 7, 8, 10, 11\}; \\ \text{NEbot}(\sigma) &= \{1, 2, 4, 5, 6, 9\} \text{ and } \text{NEtop}(\sigma) = \{3, 6, 7, 8, 9, 11\}. \end{aligned}$$

We now construct the biwords σ_{exc} and σ_{nexc}

$$\sigma_{\text{exc}} := \begin{pmatrix} \eta \\ \eta' \end{pmatrix} = \begin{pmatrix} 1_0 & 2_1 & 4_0 & 5_2 & 10_0 \\ 8 & 3 & 10 & 7 & 11 \end{pmatrix}, \quad \sigma_{\text{nexc}} := \begin{pmatrix} \zeta \\ \zeta' \end{pmatrix} = \begin{pmatrix} 3_0 & 6_2 & 7_2 & 8_1 & 9_1 & 11_0 \\ 1 & 5 & 6 & 4 & 9 & 2 \end{pmatrix}.$$

Combining these two biwords yields the permutation $\sigma = 8311075649112$; see Fig. 4.

Definition 3.4 (Indexed crossing). For $k \in [n]$ and $\sigma \in \mathfrak{S}_n$, the upper and lower crossings indexed at k of σ are defined, respectively, by

$$\text{cros}_k^+(\sigma) := \#\{l \mid l < k \leq \sigma_l < \sigma_k\} \text{ and } \text{cros}_k^-(\sigma) = \#\{l \mid \sigma_k < \sigma_l < k < l\}.$$

The number of crossings indexed at k is $\text{cros}_k(\sigma) = \text{cros}_k^+(\sigma) + \text{cros}_k^-(\sigma)$; see Fig 5.

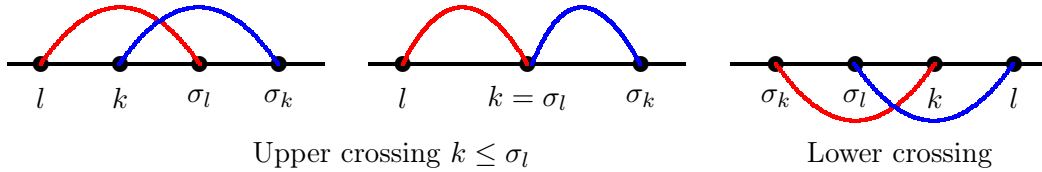


FIGURE 5. Upper crossing and lower crossing indexed at k .

Definition 3.5 (Indexed nesting). For $k \in [n]$ and $\sigma \in \mathfrak{S}_n$, the upper and lower nestings indexed at k of σ are defined, respectively, by

$$\text{nest}_k^+(\sigma) := \#\{l \mid l < k \leq \sigma_k < \sigma_l\} \text{ and } \text{nest}_k^-(\sigma) = \#\{l \mid \sigma_l < \sigma_k < k < l\}.$$

The number of crossings indexed at k is $\text{nest}_k(\sigma) = \text{nest}_k^+(\sigma) + \text{nest}_k^-(\sigma)$; see Fig. 6.

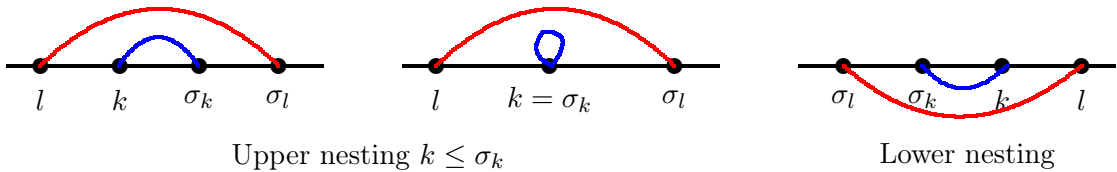


FIGURE 6. Upper nesting and lower nesting indexed at k .

Note that for any $\sigma \in \mathfrak{S}_n$, we have

$$\text{cros}(\sigma) = \sum_{k=1}^n \text{cros}_k(\sigma) \text{ and } \text{nest}(\sigma) = \sum_{k=1}^n \text{nest}_k(\sigma)$$

The definition of labels p_i in (3.14) admits a straightforward interpretation in terms of the statistic nesting,

$$p_i := \begin{cases} \text{nest}_i^+(\sigma), & \text{if } \sigma(i) \geq i \quad \text{i.e. } i \in \text{Cval}(\sigma) \cup \text{Cda}(\sigma) \cup \text{Fix}(\sigma); \\ \text{nest}_i^-(\sigma), & \text{if } \sigma(i) < i \quad \text{i.e. } i \in \text{Cpk}(\sigma) \cup \text{Cdd}(\sigma). \end{cases}$$

The following Lemma is due to Corteel [Cor07, Lemma 3]. Since the original proof is brief, we provide a visual proof based on arc diagrams of permutations.

Lemma 3.5 ([Cor07]). *Let $\Psi_{FZ}(\sigma) = (w, p) \in \mathcal{L}_n^*$ for $\sigma \in \mathfrak{S}_n$. For any $k \in [n]$, if h_k is the height of the k th step of w , then*

$$\text{cros}_k(\sigma) + \text{nest}_k(\sigma) = \begin{cases} h_k, & \text{if } k \leq \sigma(k); \\ h_k - 1, & \text{if } k > \sigma(k). \end{cases} \quad (3.18)$$

Proof. It is easy to check that the height of the k th step of w is equal to the number of upper arcs (or lower arcs) that cover node k , that is, more precisely, arcs satisfying one of the following conditions:

- the left endpoint lies to the left of k and the right endpoint lies to the right of k ;
- the left endpoint lies to the left of k and the right endpoint is k ;
- the left endpoint is k and the right endpoint lies to the right of k ;
- both the left and right endpoints are k .

In view of the upper and lower crossing (resp. nesting) indexed at k , see Definition 3.4 and 3.5, it suffices to show that

- (1) if $k \in \text{Fix}(\sigma)$, then $\text{cros}_k^+(\sigma) = 0$ and $\text{nest}_k^+(\sigma) = h_k$;
- (2) if $k \in \text{Cval}(\sigma)$, then

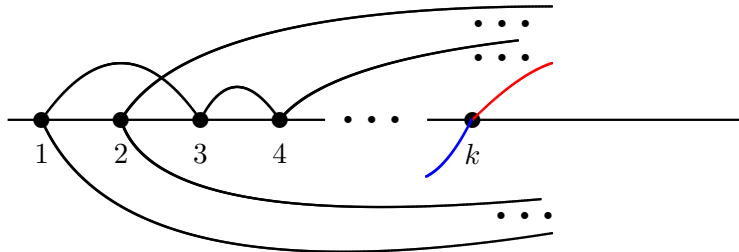
$$\text{cros}_k^+(\sigma) + \text{nest}_k^+(\sigma) = h_k \text{ with } 0 \leq \text{nest}_k^+(\sigma) \leq h_k;$$

- (3) if $k \in \text{Cda}(\sigma)$, then

$$\text{cros}_k^+(\sigma) + \text{nest}_k^+(\sigma) = h_k \text{ with } 0 \leq \text{nest}_k^+(\sigma) \leq h_k - 1;$$

- (4) and if $k \in \text{Cpk}(\sigma) \cup \text{Cdd}(\sigma)$, then

$$\text{cros}_k^-(\sigma) + \text{nest}_k^-(\sigma) = h_k - 1 \text{ with } 0 \leq \text{nest}_k^-(\sigma) \leq h_k - 1.$$



Case (1) is obvious. If $k \in \text{Cval}(\sigma)$, and there are $h_k = j$ arcs covering k , then the arc from k to $\sigma(k)$ must intersect exactly r of these arcs, for some $0 \leq r \leq j$. We order these j arcs from top to bottom and denote them by $a_1, a_2, \dots, a_j$. Suppose that the arc $(k \rightarrow \sigma(k))$ intersects exactly i of the j arcs above k , namely $a_{j-i+1}, a_{j-i}, \dots, a_j$, where $0 \leq i \leq j$. These intersected arcs together with the arc $(k \rightarrow \sigma(k))$ contribute

$$\text{cros}_k^+(\sigma) = i,$$

while the remaining $j - i$ arcs that do not intersect $(k \rightarrow \sigma(k))$ contribute

$$\text{nest}_k^+(\sigma) = j - i,$$

which proves (2).

If $k \in \text{Cda}(\sigma)$ and there are $h_k = j$ arcs that cover k , exactly one of which has right endpoint k (this arc together with the arc $(k \rightarrow \sigma(k))$ forms an upper crossing), then the arc from k to $\sigma(k)$ must intersect exactly r of the remaining $j - 1$ arcs, for some $0 \leq r \leq j - 1$. We order these $j - 1$ arcs from top to bottom and denote them by $a_1, a_2, \dots, a_{j-1}$. Suppose that the arc $(k \rightarrow \sigma(k))$ intersects exactly i of the $j - 1$ arcs above k , namely $a_{j-i}, a_{j-i-1}, \dots, a_{j-1}$, where $0 \leq i \leq j - 1$. These intersected arcs, together with the arc $(k \rightarrow \sigma(k))$, contribute

$$\text{cros}_k^+(\sigma) = i,$$

while the remaining $j - i - 1$ arcs that do not intersect $(k \rightarrow \sigma(k))$ contribute

$$\text{nest}_k^+(\sigma) = j - i - 1,$$

which proves (3).

If $k \in \text{Cpk}(\sigma) \cup \text{Cdd}(\sigma)$ and there are $h_k = j$ arcs below k , denoted by $b_1, b_2, \dots, b_j$ from bottom to top, exactly one of which has a right endpoint k , namely b_i where $1 \leq i \leq j$, then the arc $(k \rightarrow \sigma(k))$ must intersect exactly $b_{i+1}, b_{i+2}, \dots, b_j$. These intersected arcs, together with the arc $(k \rightarrow \sigma(k))$, contribute

$$\text{cros}_k^-(\sigma) = j - i,$$

while the remaining $i - 1$ arcs that do not intersect the arc $(k \rightarrow \sigma(k))$ contribute

$$\text{nest}_k^-(\sigma) = i - 1,$$

which proves (4). Eq. (3.18) follows by induction on k and can be proved analogously to Lemma 3.2; we omit the details. $\square$

Lemma 3.6. *For $\sigma \in \mathfrak{S}_n$, let $\Psi_{FZ}(\sigma) := (w, p) \in \mathcal{L}_n^*$ be the corresponding 3-restricted Laguerre history. For each $i \in [n]$, denote by h_i the height of the i th step of w . Then*

$$\text{depth}(\sigma) = \text{area}(w). \quad (3.19)$$

Proof. By Definition (2.9) and (3.15a), we see that

$$\text{depth}(\sigma) = \text{Etop}(\sigma) - \text{Ebot}(\sigma) = \text{Cpk}(\sigma) - \text{Cval}(\sigma). \quad (3.20)$$

Translating (3.20) through the Ψ_{FZ} yields

$$\text{depth}(\sigma) = \sum_{w_i=S} i - \sum_{w_j=N} j. \quad (3.21)$$

By the same argument in the proof of Lemma 3.4 we obtain (3.19). $\square$

Example 12. If $\sigma = 8311075649112 \in \mathfrak{S}_{11}$ and the corresponding Motzkin path is $w = \text{NNSNN}\mathbf{E}_1\text{SS}\mathbf{E}_3\mathbf{E}_2\text{S}$, see Fig 4, then by Definition (2.9), we have

$$\text{depth}(\sigma) = (8-1) + (3-2) + (10-4) + (7-5) + (11-10) = 17.$$

Moreover, we have

$$\text{area}(w) = \sum_{i=1}^{11} h_i = 5 \cdot 1 + 3 \cdot 2 + 2 \cdot 3 = 17.$$

3.2. Proof of Theorem 2.1 and Theorem 2.2.

Lemma 3.7. Given a permutation $\sigma \in \mathfrak{S}_{n+1}$ and $x \in [n]$, letter x is a

- left-to-right maximum of σ , if and only if p_x is equal to 0 and the x th step is S or $\mathbf{E}_2$ in the Laguerre history $\Psi_{FV}(\sigma)$;
- right-to-left maximum of σ , if and only if p_x is equal to h_x and the x th step is S or $\mathbf{E}_1$ in the Laguerre history $\Psi_{FV}(\sigma)$.

Proof. It is easy to see that if x is a left-to-right (resp. right-to-left) maximum of σ , then x is a peak or a double ascent (resp. a peak or a double descent) of σ . By the construction of Ψ_{FV} , the x th step of $\Psi_{FV}(\sigma)$ is given by S or $\mathbf{E}_2$ (resp. S or $\mathbf{E}_1$) if x is a left-to-right (resp. right-to-left) maximum of σ . If x is a left-to-right maximum of σ , i.e., there is no element to the left of x that is greater than x , which means $p_x = (31-2)_x = 0$. If x is a right-to-left maximum of σ , i.e., there is no element to the right of x that is greater than x , which is $(2-13)_x = 0$. Then by Lemma 3.2, we derive that $(31-2)_x = h_x$. $\square$

Proof of Theorem 2.1. For $\sigma \in \mathfrak{S}_{n+1}$, let $\Psi_{FV}(\sigma) = (w, p) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n$ be the corresponding Laguerre history. Recall (2.1b) that

$$\begin{aligned} \mathbf{wt}(\sigma) &= (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} \\ &\quad \times f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)}. \end{aligned} \quad (3.22)$$

Combining Lemmas 3.2–3.4 and 3.7, we have

$$\begin{aligned} \mathbf{wt}(\sigma) & h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} \\ &= \prod_{w_i=\text{N}} (u_1 h^{h_i} p^{p_i} q^{h_i-p_i}) \prod_{w_i=\text{S}} (u_2 h^{h_i} (\alpha f)^{\chi(p_i=0)} (\beta g)^{\chi(p_i=h_i)} p^{p_i} q^{h_i-p_i}) \\ &\quad \times \prod_{w_i=\mathbf{E}_1} (u_4 h^{h_i} (\beta t)^{\chi(p_i=h_i)} p^{p_i} q^{h_i-p_i}) \prod_{w_i=\mathbf{E}_2} (u_3 h^{h_i} (\alpha t)^{\chi(p_i=0)} p^{p_i} q^{h_i-p_i}). \end{aligned} \quad (3.23)$$

Summing over $p_i \in \{0, \dots, h_i\}$ for $i \in [n]$ we obtain

$$\begin{aligned} A_n(\mathbf{u}, f, g, h, t \mid \alpha, \beta; p, q) &= \sum_{w \in \mathcal{M}_n^2} \prod_{w_i=\text{N}} u_1 h^{h_i} [h_i + 1]_{p,q} \prod_{w_i=\text{S}} u_2 h^{h_i} [h_i + 1; \beta g, \alpha f]_{p,q} \\ &\quad \times \prod_{w_i=\mathbf{E}_1} u_4 h^{h_i} [h_i + 1; \beta t, 1]_{p,q} \prod_{w_i=\mathbf{E}_2} u_3 h^{h_i} [h_i + 1; 1, \alpha t]_{p,q}, \end{aligned} \quad (3.24)$$

where $\mathcal{M}_n^2$ is the set of Motzkin paths of length n with two colors in step E. This completes the proof by an application of Flajolet's fundamental Lemma 3.1. $\square$

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$, we say that an integer $i \in [n]$ is a

- left-to-right maximum index of σ , if $\sigma_j < \sigma_i$ for all $j < i$;
- right-to-left minimum index of σ , if $\sigma_j > \sigma_i$ for all $j > i$.

Since σ is a permutation, the map $i \mapsto \sigma(i) = \sigma_i$ is a bijection between positions and values. Consequently, counting left-to-right maxima (resp. right-to-left minima) by record positions or by record values gives the same cardinality. In particular,

$$\begin{aligned} \text{lmax}(\sigma) &= \#\{i \in [n] : \sigma_j < \sigma_i \text{ for all } j < i\}; \\ \text{rmin}(\sigma) &= \#\{i \in [n] : \sigma_j > \sigma_i \text{ for all } j > i\}. \end{aligned}$$

The following result provides a characterization of rmin and lmax through the bijection Ψ_{FZ} .

Lemma 3.8. *Given a permutation $\sigma \in \mathfrak{S}_n$ and $x \in [n]$, an integer x is a*

- left-to-right maximum index of σ if and only if p_x is equal to 0 and the x th step is S, E_2 , or E_3 in the 3-restricted Laguerre history $\Psi_{FZ}(\sigma)$;
- right-to-left minimum index of σ if and only if p_x is equal to 0 and the x th step is N, E_1 , or E_3 in the 3-restricted Laguerre history $\Psi_{FZ}(\sigma)$.

Proof. One can readily show that if x is a left-to-right maximum (resp. right-to-left minimum) index of σ , then $\sigma(x) \geq x$ (resp. $\sigma(x) \leq x$). By the construction of Ψ_{FZ} , the x th step of $\Psi_{FZ}(\sigma)$ is given by S, E_2 or E_3 (resp. N, E_1 or E_3) if x is a left-to-right maximum (resp. right-to-left minimum) index of σ . If x is a left-to-right maximum index of σ , then there is no upper arc above the arc $(x \rightarrow \sigma(x))$ because $(\sigma(i) < \sigma(x))$ for all $i < x$; and if x is a right-to-left minimum index of σ , then there is no lower arc below the arc $(x \rightarrow \sigma(x))$ because $(\sigma(i) > \sigma(x))$ for all $i > x$. $\square$

Lemma 3.9 ([MV94]). *Let $\sigma \in \mathfrak{S}_n$ and $\Psi_{FZ}(\sigma) = (w, p) \in \mathcal{L}_n^*$. Then*

$$\text{inv}(\sigma) = \text{area}(w) + \text{nest}(\sigma) = \sum_{i=1}^n (h_i + p_i), \quad (3.25)$$

where h_i denotes the height of the i th step of w for $i \in [n]$.

Proof of Theorem 2.2. For $\sigma \in \mathfrak{S}_n$, let $\Psi_{FZ}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n^*$ be the corresponding 3-restricted Laguerre history. Define the weight function

$$\mathbf{wt}^c(\sigma) := (u_1 u_2)^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)} \beta^{\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}. \quad (3.26)$$

Combining Lemmas 3.6, 3.8, and 3.9, we have

$$\begin{aligned} \mathbf{wt}^c(\sigma) &= \prod_{w_i=\text{N}} (u_1 \beta^{\chi(p_i=0)} p^{h_i-p_i} q^{p_i} s^{h_i+p_i} h^{h_i}) \prod_{w_i=\text{S}} (u_2 \alpha^{\chi(p_i=0)} p^{h_i-p_i-1} q^{p_i} s^{h_i+p_i} h^{h_i}) \\ &\quad \times \prod_{w_i=\text{E}_1} (u_4 \beta^{\chi(p_i=0)} p^{h_i-p_i-1} q^{p_i} s^{h_i+p_i} h^{h_i}) \prod_{w_i=\text{E}_2} (u_3 \alpha^{\chi(p_i=0)} p^{h_i-p_i} q^{p_i} s^{h_i+p_i} h^{h_i}) \\ &\quad \times \prod_{w_i=\text{E}_3} (t(\alpha\beta)^{\chi(h_i=0)} q^{h_i} s^{2h_i} h^{h_i}). \end{aligned} \quad (3.27)$$

By Definition 3.2, summing over $p_i \in \{0, 1, \dots, h_i\}$ if $w_i \in \{\text{N}, \text{E}_2, \text{E}_3\}$ and $p_i \in \{0, 1, \dots, h_i - 1\}$ if $w_i \in \{\text{S}, \text{E}_1\}$ for $i \in [n]$ we have

$$C_n(\mathbf{u}, t, s, h \mid \alpha, \beta; p, q) = \sum_{w \in \mathcal{M}_n^3} \prod_{w_i=\text{N}} u_1 (sh)^{h_i} [h_i + 1; \beta, 1]_{p,qs} \prod_{w_i=\text{S}} u_2 (sh)^{h_i} [h_i; \alpha, 1]_{p,qs}$$

$$\times \prod_{w_i=\mathbf{E}_1} u_4(sh)^{h_i}[h_i; \beta, 1]_{p,qs} \prod_{w_i=\mathbf{E}_2} u_3p(sh)^{h_i}[h_i; \alpha, 1]_{p,qs} \prod_{w_i=\mathbf{E}_3} t(\alpha\beta)^{x(h_i=0)}(qs^2h)^{h_i},$$

where $\mathcal{M}_n^3$ denotes the set of all 3-Motzkin paths of length n , that is, Motzkin paths whose horizontal step E is endowed with three possible colors: $\mathbf{E}_1$, $\mathbf{E}_2$, and $\mathbf{E}_3$. By applying Flajolet's fundamental Lemma 3.1, the proof is complete. $\square$

4. TWO BIJECTIONS OVER PERMUTATIONS

The bijections Φ_{SZ} and Φ_C actually arise from the composition of a restricted Françon–Viennot bijection and a simplified Foata–Zeilberger bijection. First, we need a variant of the bijection Ψ_{FV} (resp. Ψ_{FZ}) defined in (3.4) (resp. (3.13)) as follows.

Definition 4.1 (2-restricted Laguerre history). *A 2-restricted Laguerre history of length n is a couple (w, p) , where $w = w_1 \cdots w_n$ is a 2-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is an integer sequence satisfying $1 \leq i \leq n$,*

$$0 \leq p_i \leq \begin{cases} h_i, & \text{if } w_i \in \{N, \mathbf{E}_2\}; \\ h_i - 1, & \text{if } w_i \in \{S, \mathbf{E}_1\}, \end{cases}$$

where h_i is the height of the i th step of w . Denote by $\tilde{\mathcal{L}}_n^*$ the set of 2-restricted Laguerre histories of length n .

We describe the restricted version of Françon–Viennot's bijection $\tilde{\Psi}_{FV} : \mathfrak{S}_n \rightarrow \tilde{\mathcal{L}}_n^*$ as follows: For a permutation $\sigma \in \mathfrak{S}_n$ with a boundary condition $\sigma_0 = 0$ and $\sigma_{n+1} = n+1$, the 2-restricted Laguerre history $\tilde{\Psi}_{FV}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \tilde{\mathcal{L}}_n^*$ is defined by

$$w_i = \begin{cases} N \text{ (resp. } S) & \text{if } i \text{ is a valley (resp. peak) of } \sigma; \\ \mathbf{E}_1 \text{ (resp. } \mathbf{E}_2) & \text{if } i \text{ is a double descent (resp. double ascent) of } \sigma, \end{cases} \quad (4.1)$$

and $p_i = (31-2)_i(\sigma)$ for each $i \in [n]$.

Lemma 4.1. [Cor07] *Given a permutation $\sigma \in \mathfrak{S}_n$ and the corresponding 2-restricted Laguerre history $\tilde{\Psi}_{FV}(\sigma) = (w, p) \in \tilde{\mathcal{L}}_n^*$, for any $x \in [n]$, we have*

$$(31-2)_x(\sigma) + (2-31)_x(\sigma) = \begin{cases} h_x, & \text{if } w_x \in \{N, \mathbf{E}_2\}; \\ h_x - 1, & \text{if } w_x \in \{S, \mathbf{E}_1\}. \end{cases} \quad (4.2)$$

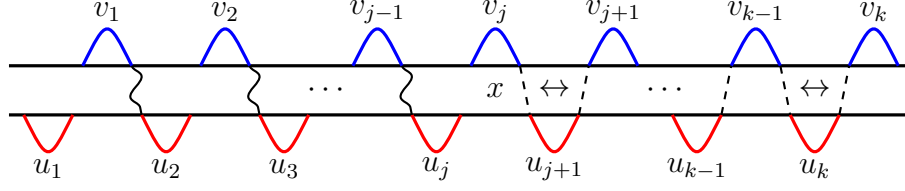
Proof. Let $\sigma \in \mathfrak{S}_n$ and $\sigma' = \sigma \star \sigma_{n+1} \in \mathfrak{S}_{n+1}$ with $\sigma_{n+1} = n+1$, where $\star$ denotes the concatenation operator of two words. Suppose that for $x \in [n]$, the x -decomposition of σ' is

$$\sigma' = u_1 v_1 \cdots u_k v_k, \quad k \geq 1,$$

where x lies inside v_j , see (3.5). It is straightforward to verify the following:

- word v_k has $n+1$ as its last element, possibly together with other entries preceding n .
- If x is a peak or a double descent of σ' , then x cannot lie in v_k .

When x is a valley or a double ascent, the word v_j must contain at least one element larger than x on the right side. Consequently, each contribution to $(2-13)_x(\sigma')$ corresponds to a contribution to $(2-31)_x(\sigma)$. When x is a peak or a double descent, and the word v_j contains x as the last element. In these two cases, one contribution to $(2-31)_x(\sigma)$ is lost. See the corresponding illustration below.



Thus, for any $x \in [n]$, we have

$$(31-2)_x(\sigma) = (31-2)_x(\sigma'), \quad (4.3a)$$

and

$$(2-31)_x(\sigma) = \begin{cases} (2-13)_x(\sigma') & \text{if } x \text{ is a valley or a double ascent;} \\ (2-13)_x(\sigma') - 1 & \text{if } x \text{ is a peak or a double descent.} \end{cases} \quad (4.3b)$$

Combining Eqs. (4.3) and Lemma 3.2, we derive (4.2). $\square$

We describe a simplified version of Foata–Zeilberger’s bijection $\tilde{\Psi}_{FZ}$ as follows: For a permutation $\sigma \in \mathfrak{S}_n$, the 2-restricted Laguerre history $\tilde{\Psi}_{FZ}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \tilde{\mathcal{L}}_n^*$ is defined by

$$w_i = \begin{cases} \mathbf{N} & \text{if } i \text{ is a cycle valley of } \sigma; \\ \mathbf{S} & \text{if } i \text{ is a cycle peak of } \sigma; \\ \mathbf{E}_1 & \text{if } i \text{ is a cycle double descent of } \sigma; \\ \mathbf{E}_2 & \text{if } i \text{ is a cycle double ascent or a fixed point of } \sigma, \end{cases} \quad (4.4)$$

and with $p_i := \text{nest}_i(\sigma)$ for each $i \in [n]$.

The following property follows directly from Lemma 3.5 and (4.4).

Lemma 4.2 ([Cor07]). *Given a permutation $\sigma \in \mathfrak{S}_n$, and the corresponding 2-restricted Laguerre history $\tilde{\Psi}_{FZ}(\sigma) = (w, p) \in \tilde{\mathcal{L}}_n^*$ for $\sigma \in \mathfrak{S}_n$. For any $k \in [n]$, if h_k is the height of the k th step of w , then*

$$\text{cros}_k(\sigma) + \text{nest}_k(\sigma) = \begin{cases} h_k, & \text{if } w_x \in \{\mathbf{N}, \mathbf{E}_2\}; \\ h_k - 1, & \text{if } w_x \in \{\mathbf{S}, \mathbf{E}_1\}. \end{cases} \quad (4.5)$$

In particular, if $k \in [n]$ is a fixed point of σ , we have

$$\text{nest}_k(\sigma) = h_k. \quad (4.6)$$

For a 2-restricted Laguerre history $(w, (p_1, \dots, p_n)) \in \tilde{\mathcal{L}}_n^*$, we define the complement operator ρ by

$$\rho((w, (p_1, \dots, p_n))) = (w, (p'_1, \dots, p'_n)), \quad (4.7)$$

where the pairs p_i and p'_i are related by

$$p_i + p'_i = \begin{cases} h_k, & \text{if } w_x \in \{\mathbf{N}, \mathbf{E}_2\}; \\ h_k - 1, & \text{if } w_x \in \{\mathbf{S}, \mathbf{E}_1\}. \end{cases} \quad (4.8)$$

4.1. **Bijection Φ_C .** We recall Corteel's bijection $\Phi_C := \tilde{\Psi}_{FZ}^{-1} \circ \tilde{\Psi}_{FV}$ in [Cor07, Sec. 5], which is constructed in such a way that if $\tau = \Phi_C(\sigma)$, then

$$(31-2)_k(\sigma) = \text{nest}_k(\tau) \quad \forall k = 1, \dots, n. \quad (4.9)$$

For $k \in [n]$, the pattern $(2-31)_k(\sigma)$ (resp. $(31-2)_k(\sigma)$) indexed by k is called the *right embracing number* (resp. *left embracing number*) of k in σ . Given a permutation σ , one constructs two biwords, $\begin{pmatrix} f \\ f' \end{pmatrix}$ and $\begin{pmatrix} g \\ g' \end{pmatrix}$, and then forms the biword

$$\tau = \begin{pmatrix} f & g \\ f' & g' \end{pmatrix}$$

by concatenating f and g , and f' and g' , respectively. These four words are defined as follows:

- The word f (resp. g) is defined as the subword of nondescent (resp. descent) tops in σ , ordered increasingly;
- The word f' (resp. g') is the permutation of nondescent (resp. descent) bottoms in σ obtained from the left-to-right (resp. right-to-left) inversion table (see Definition 3.3) whose entries are the left embracing numbers of σ .

See the following illustration:

$$\begin{pmatrix} \text{Nondescent tops (with labels)} \\ \text{Nondescent bottoms} \end{pmatrix} \xrightarrow[\text{with entries } (31-2)_i]{\text{left-to-right inversion table}} \begin{pmatrix} f \\ f' \end{pmatrix}; \quad (4.10)$$

$$\begin{pmatrix} \text{Descent tops (with labels)} \\ \text{Descent bottoms} \end{pmatrix} \xrightarrow[\text{with entries } (31-2)_i]{\text{right-to-left inversion table}} \begin{pmatrix} g \\ g' \end{pmatrix}. \quad (4.11)$$

By rearranging the columns of τ so that the top row is in increasing order, we obtain the permutation $\Phi_C(\sigma)$ as the bottom row of the rearranged biword.

Example 13. Let $\sigma = 412796583 \in \mathfrak{S}_9$ with the left embracing numbers

$$((31-2)_1, \dots, (31-2)_8) \sigma = (0, 1, 1, 0, 0, 0, 0, 1, 0).$$

Then

$$\begin{pmatrix} f \\ f' \end{pmatrix} = \begin{pmatrix} 1_0 & 2_1 & 3_1 & 5_0 & 7_0 \\ 7 & 2 & 4 & 8 & 9 \end{pmatrix}, \quad \begin{pmatrix} g \\ g' \end{pmatrix} = \begin{pmatrix} 4_0 & 6_0 & 8_1 & 9_0 \\ 1 & 3 & 6 & 5 \end{pmatrix},$$

$$\tau' = \begin{pmatrix} f & g \\ f' & g' \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 5 & 7 & 4 & 6 & 8 & 9 \\ 7 & 2 & 4 & 8 & 9 & 1 & 3 & 6 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 7 & 2 & 4 & 1 & 8 & 3 & 9 & 6 & 5 \end{pmatrix}.$$

and thus $\Phi_C(\sigma) = \tau = 724183965$. We verify that

- $\text{ndes}(\sigma) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{pmin}(\sigma) = |\{2\}| = 1$, $(31-2)(\sigma) = |\{41-2, 41-3, 96-8\}| = 3$, $(2-31)(\sigma) = |\{7-96, 4-83, 7-83, 6-83, 5-83\}| = 5$, $\text{MADL}(\sigma) = 15$, and $\text{NDdif}(\sigma) = \text{NDbot}(\sigma) - \text{NDtop}(\sigma) = 30 - 18 = 12$;
- $\text{wex}(\tau) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{fix}(\tau) = |\{2\}| = 1$, $\text{cros}(\tau) = 5$, $\text{nest}(\tau) = 3$, $\text{inv}(\tau) = 15$ and $\text{depth}(\tau) = (7-1) + (4-3) + (8-5) + (9-7) = 12$.

4.2. Bijection Φ_{SZ} . We describe a variant of the mapping $\Phi_{SZ} := \tilde{\Psi}_{FZ}^{-1} \circ \rho \circ \tilde{\Psi}_{FV}$ in [SZ10] as in the following.

Define $\tilde{\tau} = \Phi_{SZ}(\sigma)$ for $\sigma \in \mathfrak{S}_n$ such that

$$(2-31)_k(\sigma) = \text{nest}_k(\tilde{\tau}) \quad \forall k = 1, \dots, n. \quad (4.12)$$

Given a permutation $\sigma \in \mathfrak{S}_n$, we first construct two biwords, $\begin{pmatrix} f \\ \tilde{f}' \end{pmatrix}$ and $\begin{pmatrix} g \\ \tilde{g}' \end{pmatrix}$, and then form the biword

$$\tilde{\tau} = \begin{pmatrix} f & g \\ \tilde{f}' & \tilde{g}' \end{pmatrix}$$

by concatenating f and g , and $\tilde{f}'$ and $\tilde{g}'$, respectively. These four words are defined as follows:

- The word f (resp. g) is defined as the subword of nondescent (resp. descent) tops in σ , ordered increasingly;
- The word $\tilde{f}'$ (resp. $\tilde{g}'$) is the permutation of nondescent (resp. descent) bottoms in σ obtained from the left-to-right (resp. right-to-left) inversion table (see Definition 3.3) whose entries are the right embracing numbers of σ .

See the following illustration:

$$\begin{pmatrix} \text{Nondescent tops (with labels)} \\ \text{Nondescent bottoms} \end{pmatrix} \xrightarrow[\text{with entries } (2-31)_i]{\text{left-to-right inversion table}} \begin{pmatrix} f \\ \tilde{f}' \end{pmatrix}; \quad (4.13)$$

$$\begin{pmatrix} \text{Descent tops (with labels)} \\ \text{Descent bottoms} \end{pmatrix} \xrightarrow[\text{with entries } (2-31)_i]{\text{right-to-left inversion table}} \begin{pmatrix} g \\ \tilde{g}' \end{pmatrix}. \quad (4.14)$$

By rearranging the columns of $\tilde{\tau}$ so that the top row is in increasing order, we obtain the permutation $\Phi_{SZ}(\sigma)$ as the bottom row of the rearranged biword.

Example 14. Let $\sigma = 412796583 \in \mathfrak{S}_9$ with the right embracing numbers

$$((2-31)_1, \dots, (2-31)_9)\sigma = (0, 0, 0, 1, 1, 1, 2, 0, 0).$$

Then

$$\begin{pmatrix} f \\ \tilde{f}' \end{pmatrix} = \begin{pmatrix} 1_0 & 2_0 & 3_0 & 5_1 & 7_2 \\ 2 & 4 & 9 & 8 & 7 \end{pmatrix} \quad \begin{pmatrix} g \\ \tilde{g}' \end{pmatrix} = \begin{pmatrix} 4_1 & 6_1 & 8_0 & 9_0 \\ 3 & 5 & 1 & 6 \end{pmatrix},$$

$$\tilde{\tau}' = \begin{pmatrix} f & g \\ \tilde{f}' & \tilde{g}' \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 5 & 7 & 4 & 6 & 8 & 9 \\ 2 & 4 & 9 & 8 & 7 & 3 & 5 & 1 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 9 & 3 & 8 & 5 & 7 & 1 & 9 \end{pmatrix}.$$

and thus $\Phi_{SZ}(\sigma) = \tilde{\tau} = 249385716$. We verify that

- $\text{ndes}(\sigma) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{fmax}(\sigma) = |\{7\}| = 1$, $(31-2)(\sigma) = |\{41-2, 41-3, 96-8\}| = 3$, $(2-31)(\sigma) = |\{7-96, 4-83, 7-83, 6-83, 5-83\}| = 5$, $\text{MAD}(\sigma) = 17$, and $\text{NDdif}(\sigma) = \text{NDbot}(\sigma) - \text{NDtop}(\sigma) = 30 - 18 = 12$;
- $\text{wex}(\tilde{\tau}) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{fix}(\tilde{\tau}) = |\{7\}| = 1$, $\text{cros}(\tilde{\tau}) = 3$, $\text{nest}(\tilde{\tau}) = 5$, $\text{inv}(\tilde{\tau}) = 17$ and $\text{depth}(\tilde{\tau}) = (2-1) + (4-2) + (9-3) + (8-5) = 12$.

4.3. Proof of Theorem 2.3. We prove only (2.15); the proof of (2.16) is analogous and omitted.

Given $\sigma \in \mathfrak{S}_n$, let $\tau = \Phi_C(\sigma)$. It is easily checked that every column $\binom{i}{j}$ in $\binom{f}{f'}$ satisfies $i \leq j$ and every column $\binom{i}{j}$ in $\binom{g}{g'}$ satisfies $i > j$. It follows that each letter σ_i in f corresponds to a nondescent σ_i in π and also a weak excedance σ_i in τ , that is, for all $1 \leq i \leq n$,

$$\sigma_i < \sigma_{i+1} \quad (1 \leq i \leq n-1) \quad \text{or} \quad i = n \iff \sigma_i \leq \tau(\sigma_i). \quad (4.15)$$

Hence, we have $\text{ndes}(\pi) = \text{wex}(\tau)$. By definition, $(31-2)(\sigma)$ is the sum of the left embracing numbers in π and the sum of the inversions in the words f' and g' . If a pair (i, j) is an inversion in f' (resp. g'), then we have $j < i \leq \tau(i) < \tau(j)$ (resp. $\tau(j) < \tau(i) < j < i$) that is a nesting in τ . Thus, we have

$$(31-2)_k(\sigma) = \text{nest}_k(\tau), \quad \forall k \in [n]. \quad (4.16)$$

Combining (4.2) and (4.5), we derive

$$(2-31)_k(\sigma) = \text{cros}_k(\tau), \quad \forall k \in [n]. \quad (4.17)$$

By the definition of pmin (see Definition 2.2), if k is a postminimum in σ , then k is a nondescent bottom. Moreover, since k is also a right-to-left minimum, it follows that k is a nondescent top. Consequently, k belongs to both subwords f and f' . One can check that k is a postminimum of σ , if and only if $p_k = h_k$ and the k th step is E_2 in the 2-restricted Laguerre history $\tilde{\Psi}_{FV}(\sigma)$. Then, combining (4.16) and (4.6), we obtain $\tau(k) = k$.

As shown in [SZ10], the inversion number can be characterized in terms of the crossing and nesting numbers by

$$\text{inv}(\sigma) = \text{drop}(\sigma) + \text{cros}(\sigma) + 2 \text{nest}(\sigma). \quad (4.18)$$

Combining the above discussion with Eqs. (2.14b) and (4.18), we see that

$$\begin{aligned} \text{MADL}(\sigma) &= \text{des}(\sigma) + (2-31)(\sigma) + 2(31-2)(\sigma) \\ &= \text{drop}(\tau) + \text{cros}(\tau) + 2 \text{nest}(\tau) = \text{inv}(\tau). \end{aligned} \quad (4.19)$$

If $i \in [n]$ is a weak-excedance of τ , viz, $\tau(i) \geq i$, we say that $\tau(i)$ is a weak-excedance top and i a weak-excedance bottom. By (4.15), every nondescent top (resp. nondescent bottom) of σ corresponds naturally to a weak-excedance bottom (resp. weak-excedance top) of τ . Therefore, by Definitions (2.9) and (2.12), we have $\text{NDdif}(\sigma) = \text{depth}(\tau)$. $\square$

5. APPLICATIONS TO GAMMA-POSITIVITY

5.1. Proof of Theorem 2.4 and Theorem 2.5. Consider the following specialization of (2.4)

$$\mathbf{u} = (1, xu_2, u_3, xu_4) \quad \text{and} \quad \beta = \alpha, \quad (5.1)$$

we obtain

$$\begin{aligned} A_n(\mathbf{u}, f, g, h, t \mid \alpha, \alpha; p, q) &= \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} u_2^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} \\ &\quad \times t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2} p^{31-2(\sigma)} q^{2-13(\sigma)}. \end{aligned} \quad (5.2)$$

For $0 \leq k \leq \lfloor n/2 \rfloor$, let $\gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha | p, q)$ be the coefficient of u_2^k in the expansion

$$A_n(\mathbf{u}, f, g, h, t | \alpha, \alpha; p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha; p, q) u_2^k, \quad (5.3)$$

where $\mathbf{u} = (1, xu_2, u_3, xu_4)$.

Define the two special cases of (5.2):

$$A_n(x, u_2, u_3, u_4, f, g, h | p, q) := A_n(\mathbf{u}, f, g, h, 1 | 1, 1; p, q); \quad (5.4)$$

$$B_n(x, u_2, u_3, u_4, f, g, h | t, \alpha) := A_n(\mathbf{u}, f, g, h, t | \alpha, \alpha; 1, 1). \quad (5.5)$$

Lemma 5.1. *The following expansion formulas hold:*

$$A_n(x, u_2, u_3, u_4, f, g, h | p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(f, g, h | p, q) (xu_2)^k (u_3 + xu_4)^{n-2k}, \quad (5.6)$$

and

$$B_n(x, u_2, u_3, u_4, f, g, h | t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(f, g, h | t, \alpha) (xu_2)^k (u_3 + xu_4)^{n-2k}, \quad (5.7)$$

where

$$\gamma_{n,k}^a(f, g, h | p, q) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (5.8)$$

and

$$\gamma_{n,k}^b(f, g, h | t, \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (5.9)$$

with $\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}$.

Proof. Combining (5.3), (5.2) and Theorem 2.1 with (5.1), then substituting $u_2 \rightarrow \frac{u_2(u_3+xu_4)^2}{x}$ and $z \rightarrow \frac{z}{(u_3+xu_4)}$, we obtain

$$\sum_{n \geq 0} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha; p, q)}{x^k (u_3 + xu_4)^{n-2k}} u_2^k z^n = \mathcal{J} \left(z; \frac{b_k}{u_3 + xu_4}, \lambda_k \right), \quad (5.10)$$

where

$$b_k = h^k (u_3 [k+1; \alpha t, 1]_{p,q} + xu_4 [k+1; 1, \alpha t]_{p,q}), \quad (5.11)$$

$$\lambda_{k+1} = h^{2k+1} u_2 [k+1]_{p,q} [k+2; \alpha g, \alpha f]_{p,q} \quad (k \geq 0). \quad (5.12)$$

When $\alpha = t = 1$, we see that for $k \geq 0$,

$$b_k = h^k (u_3 + xu_4) [k+1]_{p,q}. \quad (5.13)$$

Thus, the coefficient of $u_2^k z^n$ in the left-hand side of (5.10) is a polynomial in f, g, h, p and q with coefficients independent of x, u_3 and u_4 . Setting $x = u_3 = 1$ and $u_4 = 0$ in (5.10), we derive that this coefficient is equal to $\gamma_{n,k}(1, 1, 0, f, g, h | 1, 1; p, q) = \gamma_{n,k}^a(f, g, h | p, q)$ by (5.8). It follows from (5.10) that

$$\frac{\gamma_{n,k}(x, u_3, u_4, f, g, h | 1, 1; p, q)}{x^k (u_3 + xu_4)^{n-2k}} = \gamma_{n,k}^a(f, g, h | p, q), \quad (5.14)$$

which is equivalent to (5.6).

When $p = q = 1$, we see that for $k \geq 0$,

$$b_k = h^k(k + \alpha t)(u_3 + xu_4). \quad (5.15)$$

Thus, the coefficient of $u_2^k z^n$ in the left-hand side of (5.10) is a polynomial in f, g, h, t and α with coefficients independent of x, u_3 and u_4 . Setting $x = u_3 = 1$ and $u_4 = 0$ in (5.10), we derive that this coefficient is equal to $\gamma_{n,k}(1, 1, 0, f, g, h | t, \alpha; 1, 1) = \gamma_{n,k}^b(f, g, h | t, \alpha)$ by (5.9). It follows from (5.10) that

$$\frac{\gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha; 1, 1)}{x^k(u_3 + xu_4)^{n-2k}} = \gamma_{n,k}^b(f, g, h | t, \alpha),$$

which is equivalent to (5.7). $\square$

Proof of Theorem 2.4. Setting $u_2 = u_3 = u_4 = 1$ in (5.6) of Lemma 5.1 yields (2.19). $\square$

Proof of Theorem 2.5. Setting $u_2 = u_3 = u_4 = 1$ in (5.7) of Lemma 5.1 yields (2.21). $\square$

5.2. Proof of Theorem 2.7. We consider the following special case of (2.11a)

$$\mathbf{u} = (x, u_2, xu_3, u_4) \quad \text{and} \quad \beta = \alpha \quad (5.16)$$

that is,

$$C_n(\mathbf{u}, t, s, h | \alpha, p, q) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} u_2^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} \\ \times h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}, \quad (5.17)$$

where $\mathbf{u} = (x, u_2, xu_3, u_4)$. For $0 \leq k \leq \lfloor (n-d)/2 \rfloor$, let $c_{n,k,d}(x, u_3, u_4, s, h | \alpha, p, q)$ be defined by the expansion

$$C_n(\mathbf{u}, t, s, h | \alpha, p, q) = \sum_{d=0}^n t^d \sum_{k=0}^{\lfloor (n-d)/2 \rfloor} c_{n,k,d}(x, u_3, u_4, s, h, \alpha | p, q) u_2^k. \quad (5.18)$$

Lemma 5.2. Let $\mathbf{u} = (x, u_2, xu_3, u_4)$. We have

$$C_n(\mathbf{u}, t, s, h | \alpha, p, q) = \sum_{d=0}^n t^d \sum_{k=0}^{\lfloor (n-d)/2 \rfloor} \gamma_{n,k,d}^c(s, h | \alpha, p, q) (xu_2)^k (u_4 + xu_3p)^{n-2k-d}, \quad (5.19)$$

where

$$\gamma_{n,k,d}^c(s, h | \alpha, p, q) = \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0, \text{fix}=d}} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}, \quad (5.20)$$

where $\mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0, \text{fix}=d} = \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0, \text{fix}(\sigma) = d \text{ and } \text{exc}(\sigma) = k\}$.

Proof. Combining (5.17), (5.18), and Theorem 2.2 with (5.16), then substituting $u_2 \rightarrow \frac{u_2(u_4 + xu_3p)^2}{x}$, $z \rightarrow \frac{z}{(u_4 + xu_3p)}$, and $t \rightarrow t(u_4 + xu_3p)$, we obtain

$$\sum_{n \geq 0} \sum_{d=0}^n \sum_{k=0}^{\lfloor (n-d)/2 \rfloor} \frac{c_{n,k,d}(x, u_3, u_4, s, h | \alpha, p, q) u_2^k}{x^k (u_4 + xu_3p)^{n-2k-d}} u_2^k t^d z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.21)$$

where

$$b_k = (sh)^k (tq^k s^k + [k; \alpha, 1]_{p,qs}), \quad \lambda_k = u_2 (sh)^{2k+1} [k; \alpha, 1]_{p,qs}^2. \quad (5.22)$$

Thus, the coefficient of $u_2^k t^d z^n$ in the left-hand side of (5.21) is a polynomial in s, h, α, p and q with coefficients independent of x, u_3 and u_4 . Setting $x = u_4 = 1$ and $u_3 = 0$ in (5.21), we derive that this coefficient is equal to $c_{n,k,d}(1, 0, 1, s, h \mid \alpha, p, q) = \gamma_{n,k,d}^c(s, h \mid \alpha, p, q)$ by (5.20). It follows from (5.21) that

$$\frac{c_{n,k,d}(x, u_3, u_4, s, h \mid \alpha, p, q) u_2^k}{x^k (u_4 + x u_3 p)^{n-2k-d}} = \gamma_{n,k,d}^c(s, h \mid \alpha, p, q), \quad (5.23)$$

which is equal to (5.19). $\square$

Proof of Theorem 2.7. Setting $u_2 = u_4 = 1$, $u_3 = 1/p$ and $t = t(1+x)$ (resp. $u_2 = u_4 = p$, $u_3 = 1$ and $t = t(1+x)$) in (5.19) of Lemma 5.2 yields Eq. (2.25) with $i = 1, 2$. $\square$

Proposition 5.3. *For $n \geq 1$, we have*

$$\sum_{\sigma \in \mathcal{D}_n} (-1)^{\text{nest}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = \sum_{j=1}^{\lfloor n/2 \rfloor} \binom{n-j-1}{j-1} h^{n-j} x^j (1+x)^{n-2j}. \quad (5.24)$$

Proof. Specializing Theorem 2.2 with $u_2 = u_3 = u_4 = \alpha = \beta = s = p = q = 1$, $q = -1$, and $t = 0$, we have

$$b_0 = 0, \quad b_1 = (1+x)h, \quad \lambda_1 = xh, \quad \text{and} \quad \lambda_2 = 0. \quad (5.25)$$

Thus

$$1 + \sum_{n \geq 1} \left(\sum_{\sigma \in \mathcal{D}_n} (-1)^{\text{nest}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} \right) z^n = \frac{1}{1 - \frac{\lambda_1 z^2}{1 - b_1 z}} = \frac{1 - (1+x)hz}{1 - (1+x)hz - xhz^2}. \quad (5.26)$$

For $n \geq 1$, one can check that

$$[z^n] \frac{1 - az}{1 - az - bz^2} = \sum_{j=1}^{\lfloor n/2 \rfloor} \binom{n-j-1}{j-1} a^{n-2j} b^j. \quad (5.27)$$

Substituting $a = (1+x)h$ and $b = xh$ in (5.27) and combining with (5.26) yields (5.24). $\square$

Remark 5.1. *As $\text{inv} = \text{nest} + \text{depth}$, Eq. (5.24) is equivalent to Theorem 5.4 (or Theorem 1.6) in [EFL26].*

5.3. Combinatorics of γ -coefficient factorization.

Proposition 5.4. *For $0 \leq k \leq \lfloor n/2 \rfloor$, there exist polynomials $E_{n,k}(h \mid p, q) \in \mathbb{N}[h, p, q]$ and $E_{n,k}(f, h \mid t, \alpha) \in \mathbb{N}[f, h, t, \alpha]$ such that*

$$\gamma_{n,k}^a(1, 1, h \mid p, q) = (p+q)^k E_{n,k}(h \mid p, q), \quad (5.28)$$

$$\gamma_{n,k}^b(f, f, h \mid t, \alpha) = 2^k E_{n,k}(f, h \mid t, \alpha). \quad (5.29)$$

Proof. For all $k \geq 1$, as

$$(p+q) \mid [k]_{p,q} [k+1]_{p,q}, \quad (5.30)$$

it follows from (5.12) that λ_k must have a factor $(p+q)$ if $\alpha = f = g = 1$. Combining this and (5.6) with $f = g = 1$, we prove (5.28).

When $p = q = 1$ and $g \rightarrow f$, it follows from (5.12) that

$$\lambda_k = h^{2k-1}k((k-1) + 2\alpha f)u_2 = 2 \cdot h^{2k-1}(k(k-1)/2 + k\alpha f)u_2 \quad (5.31)$$

for all $k \geq 1$. Combining (5.31) and (5.7) with $g = f$, we prove (5.29). $\square$

Remark 5.2. Brändén [Brä08] conjectured the divisibility (5.28) for $h = 1$; this conjecture was subsequently confirmed in [SZ12].

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	1		
2	1	$h(p+q)$	
3	1	$h(p+q)(h(p+q)+2)$	
4	1	$h(p+q)(h^2(p+q)^2 + 2h(p+q) + 3)$	$h^2(p+q)^2(h^2(p^2 + pq + q^2) + 1)$

TABLE 1. The first values of $\gamma_{n,k}^a := \gamma_{n,k}^a(1, 1, h \mid p, q)$ for $0 \leq 2k \leq n \leq 4$.

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	αt		
2	$\alpha^2 t^2$	$2\alpha h f$	
3	$\alpha^3 t^3$	$2\alpha h f(h + \alpha h t + 2\alpha t)$	
4	$\alpha^4 t^4$	$2\alpha h f((h + \alpha h t + \alpha t)^2 + 2\alpha^2 t^2)$	$4\alpha f h^2(h^2 + \alpha f(2h^2 + 1))$

TABLE 2. The first values of $\gamma_{n,k}^b := \gamma_{n,k}^b(f, f, h \mid t, \alpha)$ for $0 \leq 2k \leq n \leq 4$.

For the rest of this section, we focus on the combinatorial interpretations of the two quotients in Proposition 5.4. Following Foata-Strehl [FS74] and Hetyei-Reiner [HR98], we define the André permutations in terms of x -factorization.

Definition 5.1. Let $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$ and let $x = \sigma_i$ for $i \in [n]$. The x -factorization of σ is given by $\sigma = u \lambda(x) x \rho(x) v$, where

- (1) $\lambda(x) = \sigma_j \dots \sigma_{i-1}$ with $\sigma_l > x$ for $j \leq l \leq i-1$, $u = \sigma_1 \dots \sigma_{j-1}$, and $\sigma_{j-1} < x$.
- (2) $\rho(x) = \sigma_{i+1} \dots \sigma_k$ with $\sigma_l > x$ for $i+1 \leq l \leq k$, $v = \sigma_{k+1} \dots \sigma_n$, and $\sigma_{k+1} < x$.

Here, any of the words $u, \lambda(x), \rho(x), v$ may be empty.

Example 15. If $\sigma = 810145113612279 \in \mathfrak{S}_{12}$, then the 3-factorization of σ is

$$\underbrace{8101}_u \underbrace{4511}_{\lambda(3)} \underbrace{3}_{\rho(3)} \underbrace{612279}_v.$$

Definition 5.2 ([HR98]). Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$. Say that σ is an **André permutation of the first kind (resp. second kind)** if σ has no double descents, i.e., $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$, and each x -factorisation $u \lambda(x) x \rho(x) v$ of σ has property

- $\lambda(x) = \emptyset$ if $\rho(x) = \emptyset$,

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	1		
2	1	h	
3	1	$2h + h^2p + h^2q$	
4	1	$3h + 2h^2(p + q) + h^3(p + q)^2$	$h^2 + h^4(p^2 + pq + q^2)$

TABLE 3. The first values of $E_{n,k}(h | p, q)$ for $0 \leq 2k \leq n \leq 4$.

- $\max(\lambda(x)) < \max(\rho(x))$ (resp. $\min(\rho(x)) < \min(\lambda(x))$) if $\rho(x) \neq \emptyset$ and $\lambda(x) \neq \emptyset$.

Let $\mathcal{A}_n^1$ (resp. $\mathcal{A}_n^2$) be the set of André permutations of the first (resp. second) kind in $\mathfrak{S}_n$. It is known that [FS74] the cardinality of $\mathcal{A}_n^1$ (resp. $\mathcal{A}_n^2$) is the *Euler number* E_n , which is the n th coefficient in the Taylor expansion of $\sec(z) + \tan(z)$, namely,

$$\sum_{n \geq 0} E_n \frac{z^n}{n!} = \sec(z) + \tan(z). \quad (5.32)$$

For convenience, we define the subsets of $\mathcal{A}_n^1$ and $\mathcal{A}_n^2$, respectively

$$\mathcal{A}_{n, \text{des}=k}^1 := \{\sigma \in \mathcal{A}_n^1 : \text{des}(\sigma) = k\}; \quad (5.33)$$

$$\mathcal{A}_{n, \text{des}=k}^2 := \{\sigma \in \mathcal{A}_n^2 : \text{des}(\sigma) = k\}. \quad (5.34)$$

Theorem 5.5. *For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have*

$$E_{n,k}(h | p, q) = \sum_{\sigma \in \mathcal{A}_{n+1, \text{des}=k}^2} h^{\text{D}(\sigma)} p^{31-2(\sigma)-\text{des}(\sigma)} q^{2-13(\sigma)}. \quad (5.35)$$

By using a group action on permutations without double descents, Pan and Zeng [PZ21] proved (5.28) and (5.35) when $h = 1$. We shall extend their proof to the general case in Section 6.1.

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = +\infty$, a letter $\sigma_i \in [n]$ is said to be a

- *left-to-right minimum* (**lmin**) if $\sigma_j > \sigma_i$ for every $j < i$;
- *left-to-right-minimum-valley* (**lminval**) if σ_i is a left-to-right minimum and also a valley;
- *right-to-left-minimum-valley* (**rminval**) if σ_i is a right-to-left minimum and also a valley;
- *right-to-left-minimum-double-ascent* (**rlminda**) if it is a double ascent and also a right-to-left minimum.

Definition 5.3. *Let $\sigma = \sigma_1 \dots \sigma_n$ be a permutation in $\mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = +\infty$. Denote by $\text{Val}(\sigma)$ and $\text{Pk}(\sigma)$ the sets of valleys and peaks of σ , respectively. We define a variant of D by*

$$\tilde{\text{D}}(\sigma) := \sum_{i \in \text{Pk}(\sigma)} i - \sum_{j \in \text{Val}(\sigma) \setminus \{1\}} j. \quad (5.36)$$

Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$. The *complement* (resp. the *reverse*) of σ is defined by, respectively

$$\sigma^c = (n+1-\sigma_1)(n+1-\sigma_2) \dots (n+1-\sigma_n), \quad (5.37)$$

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	αt		
2	$\alpha^2 t^2$	$\alpha h f$	
3	$\alpha^3 t^3$	$\alpha h f (h + \alpha h t + 2\alpha t)$	
4	$\alpha^4 t^4$	$\alpha h f \left((h + \alpha h t + \alpha t)^2 + 2\alpha^2 t^2 \right)$	$\alpha h^2 f (h^2 + 2\alpha h^2 f + \alpha f)$

TABLE 4. The first values of $E_{n,k}(f, h, t, \alpha)$ for $0 \leq 2k \leq n \leq 4$.

and

$$\sigma^r = \sigma_n \sigma_{n-1} \cdots \sigma_1. \quad (5.38)$$

Example 16. Let $\sigma = 71385624 \in \mathfrak{S}_8$, then we have

$$\sigma^c = 28614375 \text{ and } \sigma^r = 42658317.$$

For convenience, let

$$\text{lmin}(\sigma) := \text{lmin}(\sigma) + \text{rmin}(\sigma) - 2; \quad (5.39a)$$

$$\text{lrmival}(\sigma) := \text{lminval}(\sigma) + \text{rminval}(\sigma) - 2; \quad (5.39b)$$

$$\text{lrmx}(\sigma) := \text{lmax}(\sigma) + \text{rmax}(\sigma) - 2; \quad (5.39c)$$

$$\text{lrmxpk}(\sigma) := \text{lmaxpk}(\sigma) + \text{rmaxpk}(\sigma) - 2. \quad (5.39d)$$

Lemma 5.6. The mapping $\eta := \sigma \mapsto (\sigma^c)^r$ is a bijection on $\mathfrak{S}_{n, \text{des}=k}^{dd=0}$ such that

$$(\text{des}, \text{lrmxpk}, \text{D}, \text{lmaxda}, \text{lrmx}) \sigma = (\text{des}, \text{lrmival}, \tilde{\text{D}}, \text{rminda}, \text{lrmx}) \eta(\sigma). \quad (5.40)$$

Lemma 5.6 yields another combinatorial interpretation for $\gamma_{n,k}^b(f, f, h | t, \alpha)$ defined in (5.9).

Proposition 5.7. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$\gamma_{n,k}^b(f, f, h | t, \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{dd=0}} f^{\text{lrmival}(\sigma)} h^{\tilde{\text{D}}(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{lrmx}(\sigma)} \quad (5.41)$$

Theorem 5.8. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$E_{n,k}(f, h | t, \alpha) = \sum_{\sigma \in \mathcal{A}_{n+1, \text{des}=k}^1} f^{\text{lrmival}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{lrmx}(\sigma)-1}. \quad (5.42)$$

By using a group action on permutations without double descents, Dong et al. [DLP25] proved (5.29) and (5.42) when $f = h = 1$. We shall extend their proof to the general case in Section 6.2.

We now derive a second combinatorial interpretation of $E_{n,k}(f, h | t, \alpha)$ from Theorem 5.8.

Theorem 5.9. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$E_{n,k}(f, h | t, \alpha) = \sum_{\sigma \in \mathcal{A}_{n+1, \text{des}=k}^1} f^{\text{lrmxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lrmx}(\sigma)-1}. \quad (5.43)$$

Proof. We define a bijection $\rho : \mathcal{A}_{n+1}^1 \rightarrow \mathcal{A}_{n+1}^1$ satisfying, for all $\sigma \in \mathcal{A}_{n+1}^1$

$$(\text{des}, \text{lrminval}, D, \text{rminda}, \text{rmin})\sigma = (\text{des}, \text{lmaxpk}, D, \text{lmaxda}, \text{lmax})\rho(\sigma). \quad (5.44)$$

Given $\sigma \in \mathcal{A}_{n+1}^1$, we define $\rho(\sigma)$ as follows. First, we decompose σ into cycles according to its right-to-left minima. Rewrite each cycle by placing its maximum element in the first position, then reorder the cycles from left to right in increasing order of their maximum elements. Erasing the cycle notation and concatenating the reordered cycles yields a permutation $\rho(\sigma)$, which is readily verified to be an André permutation of the first kind. It is clear that ρ is a bijection.

In this cycle decomposition, all descents of σ occur within cycles. Therefore, the cyclic rotation that moves the maximum element to the first position in each cycle does not change $\text{des}(\sigma)$. Moreover, ρ maps right-to-left minima to left-to-right maxima and right-to-left-minimum double ascents to left-to-right-maximum double ascents. This proves (5.44).

The desired result now follows from (5.42) and the bijection ρ . $\square$

Example 17. Let $\sigma = 561748239 \in \mathcal{A}_9^1$. According to its right-to-left minima, we decompose σ into disjoint cycles as

$$\sigma = (561)(7482)(3)(9).$$

Then we have

$$\sigma' := \rho(\sigma) = (3)(615)(8274)(9) = 361582749.$$

Moreover,

- $\text{des}(\sigma) = |\{2, 4, 6\}| = 3$, $\text{rminda}(\sigma) = |\{3\}| = 1$, and $\text{rmin}(\sigma) = |\{1, 2, 3, 9\}| = 4$;
- $\text{des}(\sigma') = |\{2, 5, 7\}| = 3$, $\text{lmaxda}(\sigma') = |\{3\}| = 1$, and $\text{lmax}(\sigma') = |\{3, 6, 8, 9\}| = 4$.

Define the following enumerative polynomials

$$\begin{aligned} E_n(x, h \mid p, q) &:= \sum_{k=0}^{\lfloor n/2 \rfloor} E_{n,k}(h \mid p, q) x^k \\ &= \sum_{\sigma \in \mathcal{A}_{n+1}^2} x^{\text{des}(\sigma)} h^{D(\sigma)} p^{31-2(\sigma)-\text{des}(\sigma)} q^{2-13(\sigma)}, \end{aligned} \quad (5.45a)$$

$$\begin{aligned} E_n(x, f, h \mid t, \alpha) &:= \sum_{k=0}^{\lfloor n/2 \rfloor} E_{n,k}(f, h, t, \alpha) x^k \\ &= \sum_{\sigma \in \mathcal{A}_{n+1}^1} x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} h^{D(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)-1}. \end{aligned} \quad (5.45b)$$

Theorem 5.10. We have

(1)

$$\sum_{n=0}^{\infty} E_n(x, h \mid p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.46)$$

where $b_k = h^k[k+1]_{p,q}$ and $\lambda_{k+1} = xh^{2k+1}[k+1]_{p,q}[k]_{p,q}/(p+q)$ for $k \geq 0$.

(2)

$$\sum_{n=0}^{\infty} E_n(x, f, h \mid t, \alpha) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.47)$$

where $b_k = h^k(k + \alpha t)$ and $\lambda_{k+1} = xh^{2k+1}k(k-1+2\alpha f)/2$ for $k \geq 0$.

Proof. Combining Eqs. (5.45a), (5.28), (2.19) and Theorem 2.1 with

$$(\mathbf{u}, f, g, h, t, \alpha, \beta, p, q) = (1, x, 1, 0, 1, 1, h, 1, 1, 1, p, q),$$

we obtain the first J -fraction (5.46).

Similarly, combining Eqs. (5.45b), (5.29), (2.21) and Theorem 2.1 with

$$(\mathbf{u}, f, g, h, t, \alpha, \beta, p, q) = (1, x, 1, 0, f, f, h, t, \alpha, \alpha, 1, 1),$$

we obtain the second J -fraction (5.47). $\square$

Remark 5.3. Han [Han20] asked for a combinatorial interpretation of (5.46) when $x = h = p = 1$, which is a q -analogue of the remarkable J -fraction for Euler numbers E_{n+1} :

$$\sum_{n=0}^{\infty} E_{n+1} z^n = \mathcal{J}\left(z; k+1, \binom{k+1}{2}\right). \quad (5.48)$$

Pan and Zeng [PZ21] established the $h = 1$ case of (5.46) shortly. The $f = h = 1$ case of (5.47) appeared in [XZ24].

6. FACTORIZATION OF γ -COEFFICIENTS VIA GROUP ACTIONS

6.1. Proof of Theorem 5.5. We use Pan-Zeng's group action [PZ21] to prove the following equivalent identity, for $0 \leq k \leq \lfloor n - 1/2 \rfloor$,

$$\sum_{\sigma \in \mathfrak{S}_n^{\text{dd}=0}, \text{des}=\mathbf{k}} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} = (p+q)^k \sum_{\sigma \in \mathcal{A}_{n, \text{des}=\mathbf{k}}^2} h^{\text{D}(\sigma)} p^{31-2(\sigma)-k} q^{2-13(\sigma)}. \quad (6.1)$$

For convenience, define the subset of $\mathfrak{S}_n$

$$\mathfrak{S}_n^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0\}. \quad (6.2)$$

For any permutation $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and $x \in [n]$, we shall identify σ with its x -factorization¹, i.e.,

$$\sigma = (w_1, w_2, x, w_4, w_5) = w_1 w_2 x w_4 w_5,$$

where w_2 (resp., w_4) is the maximal contiguous interval (possibly empty) immediately to the left (resp., right) of x whose letters are all greater than x . Let y_1 (resp. y_2) be the smallest element in w_2 (resp. w_4), i.e.,

$$y_1 := \min(w_2) \text{ and } y_2 := \min(w_4).$$

Definition 6.1. A valley x of σ is said to be

- good (resp. bad) if $y_1 > y_2$ (resp. $y_1 < y_2$);
- of type I if $\min(y_1, y_2)$ is a peak or double ascent;
- of type II if $\min(y_1, y_2)$ is a valley.

Proposition 6.1. Let $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and $x \in \text{Val}(\sigma)$ with $y = \min(y_1, y_2)$.

- (i) If x is of type I and y is a peak, then $w_4 = y$ (resp. $w_2 = y$) if $y_1 > y_2$ (resp. $y_1 < y_2$).
- (ii) If x is of type I and y is a double ascent, then $w_4 = yw_4''$ (resp. $w_2 = yw_2''$) with $w_2'', w_4'' \neq \epsilon$ if $y_1 > y_2$ (resp. $y_1 < y_2$).
- (iii) If x is of type II, then $w_4 = w_4' y w_4''$ (resp. $w_2 = w_2' y w_2''$) with $w_2', w_2'', w_4', w_4'' \neq \epsilon$ if $y_1 > y_2$ (resp. $y_1 < y_2$).

¹For convenience, we adopt a concise form that is equivalent to Definition 5.1.

Definition 6.2. For $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and each $x \in \text{Val}(\sigma)$ with $y = \min(y_1, y_2)$, we define its transform $\varphi(\sigma, x)$ as follows:

(1) If x is of type I and y is a peak, then

$$\varphi(\sigma, x) = \begin{cases} (w_1, y, x, w_2, w_5) & \text{if } y = y_2, \\ (w_1, w_4, x, y, w_5) & \text{if } y = y_1. \end{cases}$$

(2) If x is of type I and y is a double ascent, then

$$\varphi(\sigma, x) = \begin{cases} (w_1, yw_2, x, w'', w_5) & \text{if } y = y_2 \text{ and } w_4 = yw'', \\ (w_1, w'', x, yw_4, w_5) & \text{if } y = y_1 \text{ and } w_2 = yw''. \end{cases}$$

(3) If x is of type II, then

$$\varphi(\sigma, x) = \begin{cases} (w_1, w_2yw', x, w'', w_5) & \text{if } y = y_2 \text{ and } w_4 = w'yw'', \\ (w_1, w', x, w''yw_4, w_5) & \text{if } y = y_1 \text{ and } w_2 = w'yw''. \end{cases}$$

with $w', w'' \neq \epsilon$.

The three cases (i), (ii), (iii) of the transformation with $y = y_2$ are depicted in Fig. 7.

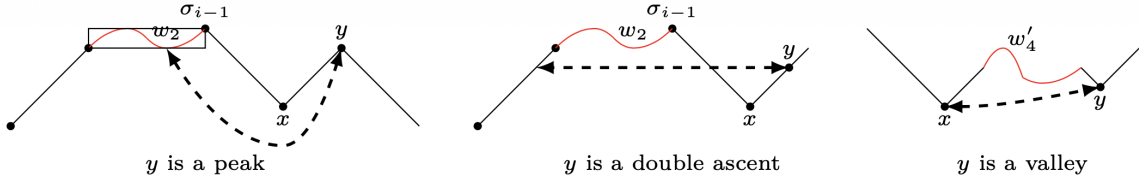


FIGURE 7. The transform $\varphi(\sigma, x)$ according to the type of y .

This transformation satisfies the following proposition.

Proposition 6.2. If $\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ and $x \in \text{Val}(\sigma)$, then $\varphi(\sigma, x) \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ and

$$D(\varphi(\sigma, x)) = D(\sigma), \quad (6.3)$$

$$(2-13) \varphi(\sigma, x) = \begin{cases} (2-13)(\sigma) + 1 & \text{if } x \text{ is good,} \\ (2-13)(\sigma) - 1 & \text{if } x \text{ is bad,} \end{cases} \quad (6.4)$$

$$(31-2) \varphi(\sigma, x) = \begin{cases} (31-2)(\sigma) - 1 & \text{if } x \text{ is good,} \\ (31-2)(\sigma) + 1 & \text{if } x \text{ is bad.} \end{cases} \quad (6.5)$$

Proof. Since the transformation $\varphi(\sigma, x)$ preserves the values at the valleys and peaks of σ , it follows directly from (2.2) that identity (6.3) holds. The proofs of (6.4) and (6.5) can be found in [PZ21, Proposition 5]. $\square$

Next, we define the transform $\varphi(\sigma, S)$ for any subset S of $\text{Val}(\sigma)$ with $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$.

Definition 6.3. Let $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$. For any $S \subseteq \text{Val}(\sigma)$, let $\{S_1, S_2\}$ be the partition of S such that

(1) S_1 is the subset of S consisting of valleys of type I, say $i_1, \dots, i_\ell$;

(2) S_2 is the subset of S consisting of valleys of type II, say $j_k < \dots < j_2 < j_1$.

Define the transforms

$$\varphi(\sigma, S_1) = \varphi(i_\ell, \dots, \varphi(i_2, \varphi(i_1, \sigma))),$$

$$\varphi(\sigma, S_2) = \varphi(j_k, \dots, \varphi(j_2, \varphi(j_1, \sigma))),$$

$$\varphi(\sigma, S) = \varphi(\varphi(\sigma, S_1), S_2).$$

Remark 6.1. Pan-Zeng prove that the image $\varphi(\sigma, S_1)$ is independent of the order of $i_1, \dots, i_\ell$, while $\varphi(\sigma, S_2)$ is defined for the elements of S_2 in the decreasing order $j_1 > j_2 > \dots > j_k$.

Proposition 6.3. If $\sigma \in \mathcal{A}_{n, \text{des}=k}^2$ and $S \subseteq \text{Val}(\sigma)$, then $\tau := \varphi(\sigma, S) \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ is well defined and

$$S = \{x \in \text{Val}(\tau) \mid x \text{ is a bad valley}\}. \quad (6.6)$$

For any set S , we denote by 2^S the set of all subsets of S . In what follows, for $\sigma \in \mathcal{A}_{n, \text{des}=k}^2$ we will identify $\text{Val}(\sigma)$ with $[k]$ under the map $a_i \mapsto i$ for $i \in [k]$ if $\text{Val}(\sigma)$ consists of $a_1 < a_2 < \dots < a_k$, and identify any subset $S \in \text{Val}(\sigma)$ with its image $S' \in 2^{[k]}$. Thus, we will use $2^{[k]}$ instead of $2^{\text{Val}(\sigma)}$.

Proposition 6.4. The map $\varphi : \mathcal{A}_{n, \text{des}=k}^2 \times 2^{[k]} \rightarrow \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ is a bijection such that for $(\sigma, S') \in \mathcal{A}_{n, k}^2 \times 2^{[k]}$ we have

$$D(\sigma) = D(\varphi(\sigma, S)); \quad (6.7)$$

$$(2\text{-}13)(\sigma) + |S'| = (2\text{-}13) \varphi(\sigma, S); \quad (6.8)$$

$$(31\text{-}2)(\sigma) - |S'| = (31\text{-}2) \varphi(\sigma, S). \quad (6.9)$$

Proof. Eq. (6.7) follows directly from (6.3). The proofs of (6.8) and (6.9) can be found in [PZ21, Proposition 7]. $\square$

Continuation of Proof of Eq. (6.1). Since $(p+q)^k = \sum_{S \in 2^{[k]}} p^{|S|} q^{k-|S|}$, it follows from Proposition 6.4 that

$$\sum_{(\sigma, S) \in \mathcal{A}_{n, \text{des}=k}^2 \times 2^{[k]}} h^{D(\sigma)} p^{31\text{-}2(\sigma) - |S|} q^{2\text{-}13(\sigma) + |S|} = \sum_{\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}} h^{D(\sigma)} p^{31\text{-}2(\sigma)} q^{2\text{-}13(\sigma)},$$

which is precisely equivalent to (6.4). $\square$

Example 18. Let $\tau = 11\mathbf{2}131\mathbf{6453897}10 \in \mathfrak{S}_{13, \text{des}=5}^{\text{dd}=0}$.

(i) First, let $(\sigma, S) := (\tau, \emptyset)$. We have $S_2 = \{1, 3\}$ and $\min(S_2) = 1$, so $S = \{1\}$ and

$$\sigma := \varphi(\sigma, 1) = 11\mathbf{1}1213\mathbf{26453897}10.$$

As $S_2 = \{3\}$, we have $\min S_2 = 3$ and $S := \{1, 3\}$, hence

$$\sigma := \varphi(\sigma, 3) = 11\mathbf{1}1213\mathbf{26354897}10.$$

(ii) Now for σ , $S_2 = \emptyset$ and $S_1 = \{4, 7\}$, then $S := \{1, 3, 4, 7\}$ and

$$\sigma := \varphi(\sigma, S_1) = 11\mathbf{1}1213\mathbf{2631078945} \in \mathcal{A}_{13, 5}^2.$$

6.2. Proof of Theorem 5.8. We prove the following equivalent identity

$$\begin{aligned} & \sum_{\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}} f^{\text{lrminval}(\sigma)} h^{\tilde{D}(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{lrmin}(\sigma)} \\ &= 2^k \sum_{\sigma \in \mathcal{A}_{n, \text{des}=k}^1} f^{\text{rminval}(\sigma)-1} h^{D(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma)-1}. \end{aligned} \quad (6.10)$$

Dong–Lin–Pan [DLP25] proved the above identity in the special case $f = h = 1$. We show that their approach extends to the general case. Recall that for any $x \in [n]$ and $\sigma \in \mathfrak{S}_n$, the x -factorization of σ is defined by

$$\sigma = w_1 w_2 x w_4 w_5,$$

where w_2 (resp., w_4) is the maximal contiguous interval (possibly empty) immediately to the left (resp., right) of x whose letters are all greater than x . Define the *swap mapping* $\varphi_x : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ by

$$\varphi_x(\sigma) := w_1 w_4 x w_2 w_5. \quad (6.11)$$

For instance, with $\sigma = 348579162 \in \mathfrak{S}_9$, in the 5-factorization of σ , $w_2 = 8$ and $w_4 = 79$ and so $\varphi_5(\sigma) = 347958162$. It is plain to see that φ_x is an involution acting on $\mathfrak{S}_n$, and for all $x, y \in [n]$, φ_x and φ_y commute. Therefore, for any $S \subseteq [n]$ we can define the function

$$\varphi_S : \mathfrak{S}_n \rightarrow \mathfrak{S}_n \quad \text{by} \quad \varphi_S = \prod_{x \in S} \varphi_x,$$

where multiplication is the composition of mappings. Hence the group $\mathbb{Z}_2^n$ acts on $\mathfrak{S}_n$ via the function φ_S . This action is known as the *Foata–Strehl action* (FS-action for short) on permutations [FS74].

Definition 6.4 (Bi-basic decomposition). *For $\sigma \in \mathfrak{S}_n$, with $\text{rmin}(\sigma) - 1 = k$ and $\text{lmin}(\sigma) - 1 = l$, there is a unique decomposition of σ as*

$$\sigma = \alpha_1 \alpha_2 \cdots \alpha_k 1 \beta_1 \beta_2 \cdots \beta_l, \quad (6.12)$$

where

- the first letter of each α_i ($1 \leq i \leq k$), which is smallest inside α_i , is a left-to-right minimum of σ ;
- and the last letter of each β_i ($1 \leq i \leq l$), which is smallest inside β_i , is a right-to-left minimum of σ .

This is the bi-basic decomposition of σ and each α_i or β_i is called a *block* of σ .

Example 19. Let $\sigma = 7\,10\,29\,5\,1\,8\,11\,3\,12\,6\,4 \in \mathfrak{S}_{12}$. Then we have

$$\textcolor{red}{7\,10} \mid \textcolor{red}{29\,5} \mid \textcolor{black}{1} \mid \textcolor{blue}{8\,11\,3} \mid \textcolor{blue}{12\,6\,4}. \quad (6.13)$$

For a word $w = w_1 w_2 \cdots w_n$, let $w^r := w_n w_{n-1} \cdots w_1$ be the reversal of w .

For a given permutation $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ with bi-basic decomposition as in (6.12) and $x \in [n]$, Dong–Lin–Pan defined the group action $\psi_x(\sigma)$ as follows.

- (1) If x is the first letter of α_i for some i , then x is a valley of σ . Let

$$\tilde{\alpha}_i := \varphi_S(\alpha_i), \quad (6.14)$$

where S is the set of all double ascents of σ inside α_i . Define $\psi_x(\sigma)$ to be the permutation obtained from σ by deleting the block α_i and then inserting $(\tilde{\alpha}_i)^r$ in some proper gap between blocks so that x becomes an additional right-to-left minimum;

- (2) If x is the last letter of β_i for some i , where β_i has length at least two², then x is also a valley of σ . Let

$$\tilde{\beta}_i := \varphi_S(\beta_i),$$

where S is the set of all double ascents of σ inside β_i . Define $\psi_x(\sigma)$ to be the permutation obtained from σ by deleting the block β_i and then inserting $(\tilde{\beta}_i)^r$ in some proper gap between blocks so that x becomes an additional left-to-right minimum.

- (3) If $x \neq 1$ is a valley of σ not in the above two cases, then set $\psi_x(\sigma) = \varphi_x(\sigma)$ defined in (6.11);
- (4) Otherwise, set $\psi_x(\sigma) = \sigma$.

As shown by Dong et al. [DLP25], for any $S \subseteq [n]$, the restriction ψ_S over $\mathfrak{S}_n^{\text{dd}=0}$ is stable and defines a $\mathbb{Z}_2^n$ -action on $\mathfrak{S}_n^{\text{dd}=0}$, which is called the *block-Foata–Strehl action* (BFS-action for short). Let $V(\sigma)$ be the set of valleys other than 1 in $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$. It is clear that $\psi_x(\sigma) = \sigma$ if and only if $x \notin V(\sigma)$.

Example 20. If $\sigma = 3485710 | 1 | 62 | 9 \in \mathfrak{S}_{10}^{\text{dd}=0}$, then $V(\sigma) = \{2, 3, 5\}$. We compute that

$$\begin{aligned} \psi_2(\sigma) &= 3485710 | 26 | 1 | 9, \\ \psi_3(\sigma) &= 1 | 62 | 4710583 | 9, \\ \psi_5(\sigma) &= 3471058 | 1 | 62 | 9, \\ \psi_{\{2,3\}}(\sigma) &= 26 | 1 | 4710583 | 9, \\ \psi_{\{2,5\}}(\sigma) &= 3471058 | 26 | 1 | 9, \\ \psi_{\{3,5\}}(\sigma) &= 1 | 62 | 4857103 | 9, \\ \psi_V(\sigma) &= 26 | 1 | 4857103 | 9. \end{aligned}$$

Lemma 6.5. For any $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and $x \in [n]$, the mapping ψ_x satisfies the following identity:

$$(\text{des}, \text{lrminval}, \tilde{D}, \text{rminda}, \text{lrmin}) \sigma = (\text{des}, \text{lrminval}, \tilde{D}, \text{rminda}, \text{lrmin}) \psi_x(\sigma). \quad (6.15)$$

Proof. From the bi-basic decomposition of σ , it is easy to see that

- the first element of each block α_i is a left-to-right-minimum-valley;
- the last element of each block β_i of length at least two is a right-to-left-minimum-valley;
- the (unique) element in each block β_i of length one is a right-to-left-minimum-double-ascent.

In the first two cases, we see that mapping ψ_x transfers a left-to-right-minimum-valley to a right-to-left-minimum-valley, or a right-to-left-minimum-valley to a left-to-right-minimum-valley. In the third case, the right-to-left-minimum-double-ascent is fixed under mapping ψ_x . Thus, the mapping ψ_x preserves the statistics lrminval , lrmin , and rminda . Moreover, since the mapping ψ_x does not change the values of the valleys and peaks of σ , it keeps $\tilde{D}$ unchanged. The reverse operator and the mapping φ_x keep des unchanged as well. Therefore, (6.15) follows. $\square$

Definition 6.5 ([HJO23]). Let $A := \{a_1, \dots, a_k\}$ be a set of k positive integers. Let $C = (a_1 \cdots a_k)$ be a cycle (cyclic permutation) of A with $a_1 = \min\{a_1, \dots, a_k\}$. Then, cycle C is called an **André cycle** if the word $a_2 \cdots a_k$ is an André permutation of the first kind. We

²In case (1), each block α_i has length at least two, as σ has no double descents.

say that a permutation is a **cycle André permutation** if it is a product of disjoint André cycles.

Let $\mathcal{CA}_n$ be the set of cycle André permutations of $[n]$. For example, $(1\ 5\ 6)(2\ 7\ 4\ 8)(3)$ is a cycle André permutation.

Lemma 6.6 ([HJO23] Theorem 4.1). *There exists a bijection $\phi : \mathcal{CA}_n \rightarrow \mathcal{A}_{n+1}^1$ as follows: let $\sigma = C_1 C_2 \dots C_l \in \mathcal{CA}_n$, where $C_1, \dots, C_l$ are André cycles ordered from left to right in increasing order of their smallest letters, which are at the end of each cycle; erasing the parentheses in σ and appending letter $n+1$ to the end results in $\phi(\sigma)$.*

Continuation of Proof of Eq. (6.10). For $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$, let $\text{Orb}(\sigma)$ be the orbit of σ under the BSF-action. As $\psi_S(\sigma) = \sigma$ if and only if $S \cap V(\sigma) = \emptyset$, we have $|\text{Orb}(\sigma)| = 2^{V(\sigma)}$. Assuming that $\text{lmin}(\sigma) = l$, then by Definition 5.2 of André permutations of the first kind, there exists a unique $\hat{\sigma} \in \text{Orb}(\sigma)$ with bi-basic decomposition

$$\hat{\sigma} = 1\beta_1\beta_2 \cdots \beta_l \quad (6.16)$$

satisfying that

($\star$) each block β_i when removing its last letter becomes an André permutation of the first kind.

By Lemma 6.5, we have

$$\sum_{\pi \in \text{Orb}(\sigma)} f^{\text{lrminal}(\pi)} h^{\tilde{D}(\pi)} t^{\text{rminda}(\pi)} \alpha^{\text{lrminal}(\pi)} = 2^{V(\hat{\sigma})} f^{\text{lrminal}(\hat{\sigma})} h^{\tilde{D}(\hat{\sigma})} t^{\text{rminda}(\hat{\sigma})} \alpha^{\text{lrminal}(\hat{\sigma})} \quad (6.17)$$

$$= 2^{\text{des}(\hat{\sigma})} f^{\text{rminval}(\hat{\sigma})-1} h^{\tilde{D}(\hat{\sigma})} t^{\text{rminda}(\hat{\sigma})} \alpha^{\text{rmin}(\hat{\sigma})-1}, \quad (6.18)$$

The second identity follows from $|V(\hat{\sigma})| = \text{des}(\hat{\sigma})$, $\text{lrminal}(\hat{\sigma}) = \text{rminval}(\hat{\sigma}) - 1$ and $\text{lrminal}(\hat{\sigma}) = \text{rmin}(\hat{\sigma}) - 1$. Summing over all orbits of $\mathfrak{S}_{n,\text{des}=k}^{\text{dd}=0}$ under the BFS-action gives

$$\sum_{\sigma \in \mathfrak{S}_{n,\text{des}=k}^{\text{dd}=0}} f^{\text{lrminal}(\sigma)} h^{\tilde{D}(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{lrminal}(\sigma)} = 2^k \sum_{\hat{\sigma} \in \mathfrak{S}_{n,\text{des}=k}^{\text{dd}=0}} f^{\text{rminval}(\hat{\sigma})-1} h^{\tilde{D}(\hat{\sigma})} t^{\text{rminda}(\hat{\sigma})} \alpha^{\text{rmin}(\hat{\sigma})-1}, \quad (6.19)$$

where

$\widehat{\mathfrak{S}}_{n,\text{des}=k}^{\text{dd}=0} := \{\hat{\sigma} \in \widehat{\mathfrak{S}}_{n,\text{des}=k}^{\text{dd}=0} : \text{bi-basic decomposition of } \hat{\sigma} \text{ has the form (6.16) satisfying } (\star)\}$.

For a word $w = w_1 w_2 \cdots w_n$, let $c(w)$ be the cyclic permutation $(w_1 - 1, w_2 - 1, \dots, w_n - 1)$. There exists a natural one-to-one correspondence

$$\hat{\sigma} = 1\beta_1\beta_2 \cdots \beta_l \mapsto c(\beta_1)c(\beta_2) \cdots c(\beta_l) = \tau. \quad (6.20)$$

between $\widehat{\mathfrak{S}}_{n,\text{des}=k}^{\text{dd}=0}$ and $\{\tau \in \mathcal{CA}_{n-1} : \text{drop}(\tau) = k\}$. Then, combining this and the bijection ϕ given in Lemma 6.6, we see $\phi(\tau) := \tau' \in \mathcal{A}_n$. One can easily check that

$$(\text{des}, \text{rminval} - 1, \tilde{D}, \text{rminda}, \text{rmin}) \hat{\sigma} = (\text{des}, \text{rminval}, D, \text{rminda}, \text{rmin}) \tau'. \quad (6.21)$$

We obtain Eq. (6.10) by combining (6.19) and (6.21). $\square$

Example 21. Let $\sigma = 5107811 | 46 | 1 | 32 | 9 \in \mathfrak{S}_{11}^{\text{dd}=0}$. Then, we have

$$\begin{aligned} \psi_5(\sigma) &= 46 | 1 | 32 | 8117105 | 9; \\ \psi_{4,5}(\sigma) &= 1 | 32 | 64 | 8117105 | 9; \\ \hat{\sigma} := \psi_{4,5,7}(\sigma) &= 1 | 32 | 64 | 1078115 | 9. \end{aligned}$$

By (6.20), we obtain

$$\hat{\sigma} \rightarrow \tau := (21)(53)(967104)(8) \in \mathcal{CA}_{10}. \quad (6.22)$$

Applying the ϕ , we have

$$\tau' := \phi(\tau) = 2153967104811 \in \mathcal{A}_{11}^1. \quad (6.23)$$

We now verify (6.21).

- On one hand, we have $\text{des}(\hat{\sigma}) = |\{2, 4, 6, 9\}| = 4$, $\text{rminval}(\hat{\sigma}) - 1 = |\{1, 2, 4, 5\}| - 1 = 3$, $\text{rminda}(\hat{\sigma}) = |\{9\}| = 1$, $\text{rmin}(\hat{\sigma}) = |\{1, 2, 4, 5, 9\}| = 5$ and $\tilde{D}(\hat{\sigma}) = 3 + 6 + 10 + 11 - (2 + 4 + 5 + 7) = 12$.
- On the other hand, we have $\text{des}(\tau') = |\{1, 3, 5, 8\}| = 4$, $\text{rminval}(\tau') = |\{1, 3, 4\}| = 3$, $\text{rminda}(\tau') = |\{8\}| = 1$, $\text{rmin}(\tau') = |\{1, 3, 4, 8, 11\}| = 5$, and $D(\tau') = 2 + 5 + 9 + 10 - (1 + 3 + 4 + 6) = 12$.

7. APPLICATION TO TOTAL-POSITIVITY

Definition 7.1 (Total positivity). A matrix of real numbers is said to be *totally positive* (TP in short) if all its minors are nonnegative.

Definition 7.2 (Coefficientwise total positivity). We say that a polynomial in $\mathbb{R}[\mathbf{X}]$, where $\mathbf{X}$ is the set of variables, is *coefficientwise-positive* if all its coefficients are nonnegative. A matrix whose entries are polynomials with real coefficients is said to be *coefficientwise-totally positive* (coefficientwise-TP, in short) if all its minors, which are polynomials themselves, are coefficientwise-positive in $\mathbf{X}$.

Let $\mathbf{a} := (a_n)_{n \geq 0}$ denote a sequence of real numbers (resp. polynomials in $\mathbb{R}[\mathbf{X}]$). To this sequence, we associate the corresponding *Hankel matrix*.

$$H_\infty(\mathbf{a}) := (a_{i+j})_{i,j \geq 0} = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots \\ a_1 & a_2 & a_3 & \cdots \\ a_2 & a_3 & a_4 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (7.1)$$

We say that the sequence $\mathbf{a}$ is *Hankel-totally positive* (resp. *coefficientwise Hankel-totally positive*) if the Hankel matrix $H_\infty(\mathbf{a})$ is totally positive (resp. coefficientwise-totally positive in $\mathbf{X}$).

7.1. S-fraction and coefficientwise-TP. For a sequence of real numbers $(a_n)_{n \geq 0}$, its Hankel matrix $(a_{i+j})_{i,j \geq 0}$ is *totally positive* if and only if there exist nonnegative real numbers $\alpha := (\alpha_n)_{n \geq 0}$ such that the following identity holds (at the level of formal power series):

$$\mathcal{S}(z; \alpha) := \frac{\alpha_0}{1 - \frac{\alpha_1 z}{1 - \frac{\alpha_2 z}{1 - \frac{\alpha_3 z}{\ddots}}}} = \sum_{n=0}^{\infty} a_n z^n. \quad (7.2)$$

This result is a combination of results due to Stieltjes [Sti94] and Gantmakher–Krein [GK37]. The continued fraction of the form on the right-hand side of equation (7.2) is called a *Stieltjes-type continued fraction* (*S-fraction* in short).

If $(a_n)_{n \geq 0}$ is instead a sequence of polynomials in one or several variables, we no longer have any equivalent condition available to us. However, if an S -fraction such as the one in equation (7.2) exists, where the $\alpha_0, \alpha_1, \dots$ are all polynomials with nonnegative coefficients, then that is a *sufficient* condition for the Hankel matrix $(a_{i+j})_{i,j \geq 0}$ to be coefficientwise-totally positive. The following fact is folklore.

Lemma 7.1. *Let $\mathbf{a} = (a_n)_{n \geq 0}$ and $\alpha = (\alpha_n)_{n \geq 0}$ be two sequences of real coefficient polynomials in $\mathbb{R}[\mathbf{X}]$ satisfying the following power series identity (7.2). If the polynomials α_n ($n \geq 0$) are coefficientwise-positive, then $\mathbf{a}$ is coefficientwise-Hankel totally positive in $\mathbf{X}$.*

This fact was first stated in this form in [Sok14], although it follows straightforwardly from the work of [Fla80] and of [Vie83, Section 3, Chapter 4]. Note that the converse of Lemma 7.1 does not hold in general.

A permutation $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$ is called an *alternating permutation* (or *zigzag permutation*) if $\sigma_1 > \sigma_2 < \sigma_3 > \sigma_4 < \cdots$. As proved by André [And79], the number of alternating permutations in $\mathfrak{S}_n$ is given by the Euler number E_n . Let $\mathcal{Z}_n$ denote the set of alternating permutations of length n . A permutation of odd length is alternating if and only if it contains neither a double ascent nor a double descent.

Define the polynomials

$$Z_n^o(x, f, g, h, \alpha, \beta | p, q) := \sum_{\sigma \in \mathcal{Z}_{2n+1}} x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} \times \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} p^{31-2(\sigma)} q^{2-13(\sigma)}. \quad (7.3)$$

Theorem 7.2. *The sequence $\mathcal{Z}_o := (Z_n(x, f, g, h, \alpha, \beta | p, q))_{n \geq 0}$ is coefficientwise-Hankel totally positive in $\mathbf{X} := \{x, f, g, h, \alpha, \beta, p, q\}$.*

Proof. By setting $\mathbf{u} = (1, x, 0, 0)$ in Theorem 2.1, we derive

$$\sum_{n=0}^{\infty} Z_n(x, f, g, h, \alpha, \beta | p, q) z^n = \mathcal{J}(\sqrt{z}; \mathbf{0}, \boldsymbol{\lambda}) = \mathcal{S}(z; \boldsymbol{\lambda}), \quad (7.4)$$

where $\lambda_k = x h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}$.

Hence, every λ_k is coefficientwise-positive in the variables $x, f, g, h, \alpha, \beta, p, q$. By Lemma 7.1, the sequence $\mathcal{Z}_o$ is therefore coefficientwise Hankel-totally positive. $\square$

A permutation σ of even length $2n$ is an alternating permutation if and only if $\sigma' := \sigma \star (2n+1)$ is an alternating permutation, where $\star$ denotes the concatenation of words.

Lemma 7.3. *Let $\sigma' := \sigma \star (2n+1) \in \mathcal{Z}_{2n+1}$ with $\sigma \in \mathcal{Z}_{2n}$. We have*

$$(\text{des}, \text{lmaxpk}, \text{D} + 2n - \sigma_{2n}, \text{lmax}) \sigma = (\text{des}, \text{lmaxpk} - 1, \text{D}, \text{lmax} - 1) \sigma'. \quad (7.5)$$

For each $x \in [2n]$, we have

$$(31-2)_x(\sigma) = (31-2)_x(\sigma') \text{ and } (2-31)_x(\sigma) = \begin{cases} (2-13)_x(\sigma') & \text{if } \sigma^{-1}(x) \text{ is even;} \\ (2-13)_x(\sigma') - 1 & \text{if } \sigma^{-1}(x) \text{ is odd.} \end{cases}$$

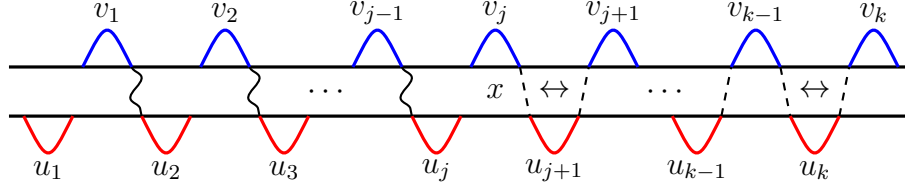
Proof. Eq. (7.5) follows directly from the Definitions of these permutation statistics. Let $\sigma' = \sigma \star (n+1) \in \mathcal{Z}_{2n+1}$, where $\sigma \in \mathcal{Z}_{2n}$. By the x -decomposition of σ' , the first equality is obvious. Suppose that for $x \in [2n]$, the x -decomposition of σ' is

$$\sigma' = u_1 v_1 \cdots u_k v_k, \quad k \geq 1,$$

where x lies in v_j , see (3.5). It is straightforward to verify the following:

- word v_k has $2n + 1$ as its last letter, possibly together with other entries preceding $2n + 1$.
- If x is a peak of σ' , then x cannot lie in v_k .

When $\sigma^{-1}(x)$ is even, i.e., x is a valley of σ' , the word v_j must contain elements that are larger than x on both sides. Consequently, each contribution to $(2-13)_x(\sigma')$ corresponds to a contribution to $(2-31)_x(\sigma)$. When $\sigma^{-1}(x)$ is odd, the letter x is a peak of σ' , and the word v_j contains only x . In this case, one contribution to $(2-31)_x(\sigma)$ is lost. See the corresponding illustration below.



This completes the proof. $\square$

Define the polynomials

$$Z_n^e(x, f, h, \alpha | p, q) := \sum_{\sigma \in \mathcal{Z}_{2n}} x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)} h^{\text{D}(\sigma) + 2n - \sigma_{2n}} \alpha^{\text{lmax}(\sigma)} p^{31-2(\sigma)} q^{2-31(\sigma)}. \quad (7.6)$$

Theorem 7.4. *The sequence $\mathcal{Z}_e := (Z_n^e(x, f, h, \alpha | p, q))_{n \geq 0}$ is coefficientwise-Hankel totally positive in $\mathbf{X} := \{x, f, h, \alpha, p, q\}$.*

Proof. Combining Lemma 7.3 and Theorem 2.1 with $\mathbf{u} = (1, x, 0, 0)$ and $\beta = 0$ we derive

$$\sum_{n=0}^{\infty} Z_n^e(x, f, h, \alpha | p, q) z^n = \mathcal{J}(\sqrt{z}; \mathbf{0}, \boldsymbol{\lambda}) = \mathcal{S}(z; \boldsymbol{\lambda}), \quad (7.7)$$

where $\lambda_k = x h^{2k-1} [k]_{p,q} [k+1; 0, \alpha f]_{p,q}$. Hence, every λ_k is coefficientwise-positive in the variables x, f, h, α, p, q . By Lemma 7.1, the sequence $\mathcal{Z}_e$ is therefore coefficientwise Hankel-totally positive. $\square$

7.2. J-fraction and coefficientwise-TP. Let $\mathbf{a} := (a_n)_{n \geq 0}$ be a sequence of polynomials in $\mathbb{R}[\mathbf{X}]$, where $\mathbf{X}$ is the set of variables. Assume that there exists a J -fraction for which the following identity is satisfied:

$$\sum_{n \geq 0} a_n t^n = \mathcal{J}(t; \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{1 - b_0 t - \frac{\lambda_1 t^2}{1 - b_1 t - \frac{\lambda_2 t^2}{1 - b_2 t - \ddots}}}. \quad (7.8)$$

Define the associated tridiagonal matrix, called *Jacobi matrix* or *production matrix* see [PSZ23],

$$M_{\infty}(\mathbf{b}, \boldsymbol{\lambda}) = \begin{pmatrix} b_0 & 1 & 0 & 0 & \dots \\ \lambda_1 & b_1 & 1 & 0 & \dots \\ 0 & \lambda_2 & b_2 & 1 & \dots \\ 0 & 0 & \lambda_3 & b_3 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (7.9)$$

The following characterization for coefficientwise-TP is due to Sokal [Sok14], see also [PSZ23, Theorem 9.7].

Theorem 7.5 ([Sok14]). *Let $\mathbf{a} := (a_n)_{n \geq 0}$ be a sequence of polynomials in $\mathbb{R}[\mathbf{X}]$ and the associated Hankel matrix $H_\infty(\mathbf{a})$. If $M_\infty(\mathbf{b}, \boldsymbol{\lambda})$ is coefficientwise-TP in $\mathbf{X}$, then so is $H_\infty(\mathbf{a})$.*

The classical criterion in [Pin10, Theorem 4.3] for tridiagonal (Jacobi) matrices over $\mathbb{R}$ extends verbatim to matrices with entries in an arbitrary partially ordered commutative ring; we shall therefore work in this generality throughout.

Lemma 7.6. *A Jacobi matrix is coefficientwise-totally positive in $\mathbf{X}$ if and only if all its off-diagonal elements and all its principal minors containing consecutive rows and columns are coefficientwise-positive in $\mathbf{X}$.*

Proof. Let $A = M_\infty(\mathbf{b}, \boldsymbol{\lambda}) = (a_{i,j})_{i,j \geq 0}$. Then $a_{i,j} = 0$ if $|i - j| > 1$. Fix $r \geq 1$ and index sets $I = (i_1 < \dots < i_r)$ and $J = (j_1 < \dots < j_r)$, and consider the minor

$$A \begin{pmatrix} I \\ J \end{pmatrix} := \det(a_{i_p, j_q})_{1 \leq p, q \leq r}.$$

If $|i_k - j_k| \geq 2$ for some k , then $A \begin{pmatrix} I \\ J \end{pmatrix} = 0$ since A is tridiagonal. Thus assume $|i_k - j_k| \leq 1$ for all k . Whenever $|i_\ell - j_\ell| = 1$, the tridiagonal form forces a block-triangular decomposition of the corresponding submatrix, yielding the factorization

$$A \begin{pmatrix} i_1, \dots, i_r \\ j_1, \dots, j_r \end{pmatrix} = A \begin{pmatrix} i_1, \dots, i_{\ell-1} \\ j_1, \dots, j_{\ell-1} \end{pmatrix} A \begin{pmatrix} i_\ell \\ j_\ell \end{pmatrix} A \begin{pmatrix} i_{\ell+1}, \dots, i_r \\ j_{\ell+1}, \dots, j_r \end{pmatrix}. \quad (7.10)$$

Iterating (7.10) we can factor any minor of A , namely, if

$$i_1 = j_1, \dots, i_{k_1} = j_{k_1}, \quad i_{k_1+1} \neq j_{k_1+1}, \dots, i_{k_2} \neq j_{k_2}, \quad i_{k_2+1} = j_{k_2+1}, \dots, i_{k_3} = j_{k_3}, \dots,$$

then

$$A \begin{pmatrix} i_1, \dots, i_r \\ j_1, \dots, j_r \end{pmatrix} = A \begin{pmatrix} i_1, \dots, i_{k_1} \\ j_1, \dots, j_{k_1} \end{pmatrix} A \begin{pmatrix} i_{k_1+1} \\ j_{k_1+1} \end{pmatrix} \dots A \begin{pmatrix} i_{k_2} \\ j_{k_2} \end{pmatrix} A \begin{pmatrix} i_{k_2+1}, \dots, i_{k_3} \\ j_{k_2+1}, \dots, j_{k_3} \end{pmatrix} \dots, \quad (7.11)$$

where

$$A \begin{pmatrix} i_k \\ j_k \end{pmatrix} = \begin{cases} 1, & \text{if } j_k = i_k + 1, \\ \lambda_{i_k}, & \text{if } j_k = i_k - 1. \end{cases}$$

Finally, if $k_t + 1 < k_{t+1}$ for some $t \in [r - 1]$, then

$$A \begin{pmatrix} k_1, \dots, k_s \\ k_1, \dots, k_s \end{pmatrix} = A \begin{pmatrix} k_1, \dots, k_t \\ k_1, \dots, k_t \end{pmatrix} A \begin{pmatrix} k_{t+1}, \dots, k_s \\ k_{t+1}, \dots, k_s \end{pmatrix}. \quad (7.12)$$

Iterating (7.12), every principal minor factors into a product of consecutive principal minors. Since coefficientwise-positive polynomials are closed under multiplication, all factors on the right-hand side of (7.11) are coefficientwise-positive, and therefore $A \begin{pmatrix} I \\ J \end{pmatrix}$ is coefficientwise-positive for all I, J . Hence A is coefficientwise-totally positive. $\square$

Define the multivariable Eulerian polynomials

$$\mathcal{A}_n(x, h | p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}. \quad (7.13)$$

Theorem 7.7. *The sequence $\mathcal{A} := (\mathcal{A}_n(x, h | p, q))_{n \geq 0}$ is coefficientwise-Hankel totally positive in $\mathbf{X} := \{x, h, p, q\}$.*

Proof. By Theorem 2.1 with $u_2 = u_4 = x$ and $u_1 = u_3 = f = g = t = \alpha = \beta = 1$, we have

$$\sum_{n \geq 0}^{\infty} A_n(x, h \mid p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (7.14)$$

where $b_k = h^k(1+x)[k+1]_{p,q}$ and $\lambda_k = xh^{2k-1}[k]_{p,q}[k+1]_{p,q}$.

Let $A = M_{\infty}(\mathbf{b}, \boldsymbol{\lambda}) = (a_{i,j})_{i,j \geq 0}$. For $k \geq 0$ and $n \geq 1$, define the contiguous principal minors

$$D_{k,n} := A \begin{pmatrix} I \\ J \end{pmatrix} \text{ with } I = J = \{k, \dots, k+n-1\}. \quad (7.15)$$

Then we have the standard continuant recurrence

$$D_{k,n} = b_{k+n-1}D_{k,n-1} - \lambda_{k+n-1}D_{k,n-2}, \quad (7.16)$$

where $D_{k,0} = 1$ and $D_{k,1} = b_k$.

We prove by induction on $n \geq 1$ that

$$D_{k,n} = h^{nk + \binom{n}{2}} (1 + x + x^2 + \dots + x^n) \left(\prod_{j=1}^n [k+j]_{p,q} \right). \quad (7.17)$$

Set $[n]_x := 1 + x + \dots + x^{n-1}$ and

$$F_{k,n} := h^{nk + \binom{n}{2}} [n+1]_x \left(\prod_{j=1}^n [k+j]_{p,q} \right).$$

Clearly $F_{k,0} = 1 = D_{k,0}$ and $F_{k,1} = (1+x)h^k[k+1]_{p,q} = b_k = D_{k,1}$. By (7.14), for $n \geq 2$, we have

$$\begin{aligned} b_{k+n-1} &= (1+x)h^{k+n-1}[k+n]_{p,q}, \\ \lambda_{k+n-1} &= xh^{2k+2n-3}[k+n-1]_{p,q}[k+n]_{p,q}, \end{aligned}$$

and a direct computation gives

$$b_{k+n-1}F_{k,n-1} - \lambda_{k+n-1}F_{k,n-2} = h^{nk + \binom{n}{2}} ((1+x)[n]_x - x[n-1]_x) \left(\prod_{j=1}^n [k+j]_{p,q} \right).$$

Since $(1+x)[n]_x - x[n-1]_x = [n+1]_x$, we obtain

$$b_{k+n-1}F_{k,n-1} - \lambda_{k+n-1}F_{k,n-2} = F_{k,n}.$$

Thus $F_{k,n}$ satisfies the same recurrence and initial conditions as $D_{k,n}$ for all $k \geq 0$ and $n \geq 1$. This establishes (7.17).

By Lemma 7.6 and Theorem 7.5, the sequence $\mathcal{A}$ is therefore coefficientwise Hankel-totally positive. □

8. CONCLUDING REMARKS

The following identities are direct consequences of the continued fraction: specializing one of the parameters to -1 makes the expression collapse into a simple closed form.

Proposition 8.1. *The following identities hold.*

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{inv}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = (1 - xh)^{n-1} \quad (8.1a)$$

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{nest}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = (1 + xh)^{n-1} \quad (8.1b)$$

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{rmin}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = (-1)^n (1 - xh)^{n-1} \quad (8.1c)$$

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{lmax}(\sigma)} h^{\text{depth}(\sigma)} = (-1)^n (1 - h)^{n-1}. \quad (8.1d)$$

Proof. If $\lambda_2 = 0$, the J -fraction (2.6) reduces to the rational fraction:

$$\mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z}} = \frac{1 - b_1 z}{(1 - b_0 z)(1 - b_1 z) - \lambda_1 z^2}. \quad (8.2)$$

Substituting $\mathbf{u} = (x, 1, x, 1)$, $p = q = t = \alpha = \beta = 1$, and $s = -1$ in Theorem 2.2, we obtain $b_0 = 1$, $b_1 = -xh$, $\lambda_1 = -xh$, $\lambda_2 = 0$, and

$$\sum_{n \geq 0} \sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{inv}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} z^n = 1 + \frac{z}{1 - (1 - xh)z}, \quad (8.3)$$

which implies (8.1a). The remaining identities follow similarly. $\square$

Remark 8.1. *Eqs. (8.1a)–(8.1c) were proved by Eu et al. [EFL26] without using continued fractions.*

Eu et al. [EFL26] considered a variant of the notions of crossing and nesting for permutations as follows:

$$\text{cros}^*(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j < \sigma_i < \sigma_j) \vee (\sigma_i < \sigma_j \leq i < j)\}; \quad (8.4a)$$

$$\text{nest}^*(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j < \sigma_j < \sigma_i) \vee (\sigma_j < \sigma_i \leq i < j)\}. \quad (8.4b)$$

We note that these statistics are closely related to cros and nest (see (2.10)) by the inverse mapping. More precisely, we have the following result.

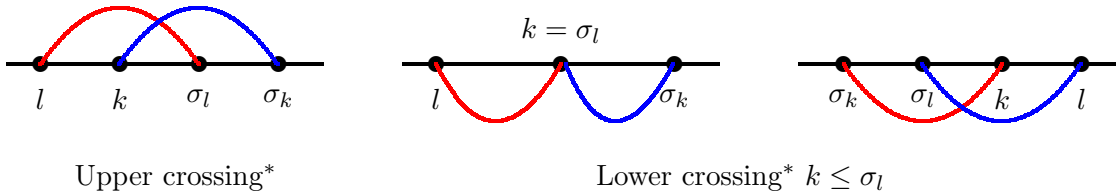
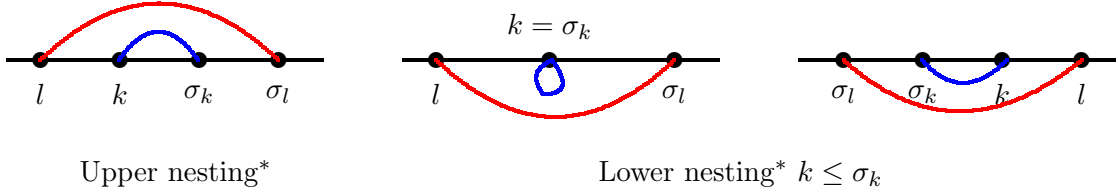


FIGURE 8. Upper crossing* and lower crossing* indexed at k .

Lemma 8.2. *The inverse mapping $\iota : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$, $\iota(\sigma) = \sigma^{-1}$ is a bijection and satisfies*

$$(\text{exc}, \text{drop}, \text{fix}, \text{depth}, \text{cros}, \text{nest}) \sigma = (\text{drop}, \text{exc}, \text{fix}, \text{depth}, \text{cros}^*, \text{nest}^*) \iota(\sigma). \quad (8.5)$$

FIGURE 9. Upper nesting* and lower nesting* indexed at k

Proof. We only verify the statistic depth; the remaining correspondences are straightforward. By Definition 2.9, we see that

$$\text{depth}(\sigma^{-1}) = \sum_{\sigma^{-1}(k) > k} (\sigma^{-1}(k) - k) = \sum_{i > \sigma(i)} (i - \sigma(i)),$$

where the second equality follows by setting $i = \sigma^{-1}(k)$. Moreover,

$$0 = \sum_{i=1}^n (\sigma(i) - i) = \sum_{\sigma(i) > i} (\sigma(i) - i) + \sum_{i > \sigma(i)} (i - \sigma(i)).$$

Combining the above two identities we obtain $\text{depth}(\sigma^{-1}) = \text{depth}(\sigma)$. □

Substituting $\mathbf{u} = (x, 1, x, 1)$ and $s = \alpha = \beta = 1$ in Theorem 2.2 we obtain

$$1 + \sum_{n \geq 1} \left(\sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} t^{\text{fix}(\sigma)} h^{\text{depth}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)} \right) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (8.6)$$

where $b_k = h^k(tq^k + (1 + xp)[k]_{p,q})$ and $\lambda_k = xh^{2k-1}[k]_{p,q}^2$.

By Lemma 8.2, since $n = \text{exc}(\sigma) + \text{fix}(\sigma) + \text{drop}(\sigma)$ for $\sigma \in \mathfrak{S}_n$, substituting $(x, t) \rightarrow (\frac{1}{x}, \frac{t}{x})$ and then $z \rightarrow xz$ in (8.6), we obtain

$$\sum_{n \geq 0} \left(\sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} t^{\text{fix}(\sigma)} p^{\text{cros}^*(\sigma)} q^{\text{nest}^*(\sigma)} h^{\text{depth}(\sigma)} \right) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (8.7)$$

where $b_k = h^k(x[k]_{p,q} + tq^k + p[k]_{p,q})$ and $\lambda_k = xh^{2k-1}[k]_{p,q}^2$.

When $t = 1$, the above J -fraction reduces to

$$\sum_{n \geq 0} \left(\sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} p^{\text{cros}^*(\sigma)} q^{\text{nest}^*(\sigma)} h^{\text{depth}(\sigma)} \right) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (8.8)$$

where $b_k = h^k(x[k]_{p,q} + [k+1]_{p,q})$ and $\lambda_k = xh^{2k-1}[k]_{p,q}^2$. This J -fraction was obtained by Eu et al. [EFL26, Theorem 1.2]. They further observed that it exhibits a symmetry $\text{nest}^* \leftrightarrow \text{cros}^*$ for fixed values of exc and depth , and asked for a bijective proof of this symmetry. The following theorem provides such a bijection.

Theorem 8.3. *Let $\iota : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$, $\iota(\sigma) = \sigma^{-1}$ and $\Phi^* = \Phi_{SZ} \circ \Phi_C^{-1}$. Then the mapping $\Theta := \iota \circ \Phi^* \circ \iota : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ is a bijection such that for any $\sigma \in \mathfrak{S}_n$,*

$$(\text{exc}, \text{nest}^*, \text{cros}^*, \text{depth}) \sigma = (\text{exc}, \text{cros}^*, \text{nest}^*, \text{depth}) \Theta(\sigma) \quad (8.9)$$

Proof. By (2.17) of Theorem 2.3 and $n = \text{wex} + \text{drop}$, the bijection Φ^* satisfies

$$(\text{drop}, \text{nest}, \text{cros}, \text{depth}) \sigma = (\text{drop}, \text{cros}, \text{nest}, \text{depth}) \Phi^*(\sigma). \quad (8.10)$$

Combining (8.5) and (8.10), we obtain (8.9). $\square$

From the construction of the bijections Φ_{SZ} and Φ_C , we see that the two bijections satisfy the following identity

$$\begin{aligned} \Phi_{SZ} \circ \Phi_C^{-1} &= \tilde{\Psi}_{FZ}^{-1} \circ \rho \circ \tilde{\Psi}_{FV} \circ (\tilde{\Psi}_{FZ}^{-1} \circ \tilde{\Psi}_{FV})^{-1} \\ &= \tilde{\Psi}_{FZ}^{-1} \circ \rho \circ \tilde{\Psi}_{FZ}, \end{aligned} \quad (8.11)$$

where $\tilde{\Psi}_{FZ}$ and ρ are defined in (4.4) and (4.7).

Besides the bijections Φ_C and Φ_{SZ} , Steingrímsson and Williams [SW07] constructed another variant Φ_{SW} of Φ_{CSZ} satisfying

$$(\text{ndes}, 31-2, 2-31) \sigma = (\text{drop} + 1, \text{nest}, \text{cros}) \Phi_{SW}(\sigma) \quad \text{for } \sigma \in \mathfrak{S}_n. \quad (8.12)$$

Let $\text{Ddif}(\sigma) := \text{Destop}(\sigma) - \text{Desbot}(\sigma)$ for $\sigma \in \mathfrak{S}_n$. It is known [CSZ97, Proposition 3] that the mapping Φ_{CSZ} implies that the two bi-statistics

$$(\text{des}, \text{Ddif}) \quad \text{and} \quad (\text{exc}, \text{depth})$$

are equidistributed on $\mathfrak{S}_n$. Two alternative proofs of this result were given by Reifegerste [Rei02] and Petersen and Tenner [PT15] using variants of the *fundamental transformation*.

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